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On the k-Numerical Range of Cyclic Shift Matrices

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Abstract

On the k-Numerical Range of Cyclic Shift Matrices

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June 2025

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In 2023, M.-T. Chien, S. Kirkland, C.-K. Li, and H. Nakazato conjectured that the cyclic shift $S(a_1, a_3, a_5, \ldots, a_6, a_4, a_2)$ with $0 \leq a_1 \leq \cdots \leq a_n$ has the largest numerical range among all cyclic shift matrices with the same collection of weights. H.-L. Gau proved the conjecture in 2024. In this study, a version of the problem for the k-numerical range is formulated. Aspects of the conjecture are explored in this new setting. For small n (at least for $n = 2, 3, 4$), a necessary and sufficient condition is established for an n-by-n cyclic shift to have the largest k-numerical range among all n-by-n cyclic shift matrices with the same collection of nonnegative real weights. The conjecture is proved for n-by-n cyclic shift matrices. The main results are extended to n-by-n cyclic shift matrices whose weights are complex. In order to prove the main results, properties of the k-numerical range are revisited. In particular, it is shown that the k-numerical range is the intersection of certain closed half-planes.

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Chapter 1

Introduction

The theory and applications of quadratic forms comprise the theory of *numerical ranges.* The numerical range of an n-by-n matrix A is defined as

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n \text{ is a unit vector}\}.$$

The numerical range can be used to estimate the eigenvalues of A through its spectral containment property [11, Property 1.2.6]. Applications can be found in quantum information science [3], dilation theory [5], and matrix polynomial factorization [16]. The numerical range has been generalized in different contexts: k-numerical range, c-numerical range, and C-numerical range [14]. In what follows, we consider the k-numerical range. For $k \in \{1, \ldots, n\}$, the *k-numerical range* of an n-by-n matrix A is defined as

$$W_k(A) = \left\{ \sum_{j=1}^{k} x_j^* A x_j : \{x_1, \ldots, x_k\} \subset \mathbb{C}^n \text{ is orthonormal} \right\}.$$

In general, $W(A)$ is a convex set due to the Toeplitz-Hausdorff theorem [11, Section 1.3]. A basic problem in the theory of numerical ranges involves further identifying properties of $W(A)$ beside its convexity. One approach is to examine numerical ranges of special classes of matrices. This study focuses on the numerical ranges of *cyclic shift* matrices. A cyclic shift matrix is a matrix of the form

$$A = \begin{bmatrix} 0 & a_1 & 0 & \cdots & 0 \\ 0 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & a_{n-1} \\ a_n & 0 & 0 & \cdots & 0 \end{bmatrix},$$

where $a_1, \ldots, a_n$ are complex numbers called the *weights* of A. We use the notation $A = S(a_1, \ldots, a_n)$ to say that A is a cyclic shift with weights ordered as $a_1, \ldots, a_n$. Cyclic shift matrices are regarded as tools in the translation matching problem [8]. Moreover, properties of cyclic shift matrices have notable connections to graph theory: for any cyclic shift A with positive real weights, $A + A^\top$ is the adjacency matrix of a weighted undirected cycle [4].

In 2023, M.-T. Chien, S. Kirkland, C.-K. Li, and H. Nakazato conjectured that the cyclic shift $S(a_1, a_3, a_5, \ldots, a_6, a_4, a_2)$ with $0 \leq a_1 \leq \cdots \leq a_n$ has the largest numerical range among all cyclic shift matrices with the same collection of weights [4]. By using a characterization of the numerical range as the intersection of a certain family of closed half-planes, they showed that the conjecture holds for $n = 2, 3, 4, 5, 6$, while claiming to have a symbolic proof for $n = 7, 8$. In 2024, the conjecture was finally proved by H.-L. Gau [7]. By applying the Rayleigh quotient theorem [12, Theorem 4.2.2], Gau extended the findings of Chien et al. to cyclic shift matrices of any size n.

In this study, we formulate a version of the problem for the k-numerical range. We explore aspects of the conjecture in this new setting. For small n (at least for $n = 2, 3, 4$), we consider the following problems:

(i) we establish a necessary and sufficient condition for an n-by-n cyclic shift to have the largest k-numerical range among all n-by-n cyclic shift matrices with the same collection of nonnegative real weights;

(ii) we prove the conjecture for n-by-n cyclic shift matrices;

(iii) we extend the main results to n-by-n cyclic shift matrices whose weights are complex.

To prove the main results, we revisit properties of the k-numerical range. In particular, we show that the k-numerical range is the intersection of certain closed half-planes.

Chapter 2

Preliminaries

In this chapter, we state some facts from linear algebra, matrix analysis, and topology to be used in the study of k-numerical ranges and cyclic shift matrices.

2.1 Some Notations

Let $\mathbb{N}$ denote the set of all natural numbers, $\mathbb{R}$ the set of all real numbers, and $\mathbb{C}$ the set of all complex numbers. For each $a+bi \in \mathbb{C}$ where $a, b \in \mathbb{R}$, its *conjugate* is given by $\overline{a+bi} = a-bi$, and its *modulus* by $|a+bi| = \sqrt{a^2+b^2}$.

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. For $m, n \in \mathbb{N}$, the set $M_{m,n}(\mathbb{F})$ denotes the collection of all m-by-n matrices over $\mathbb{F}$. For brevity, we write $M_n(\mathbb{F}) := M_{n,n}(\mathbb{F})$ and $\mathbb{F}^n := M_{n,1}(\mathbb{F})$. Denote the zero matrix in $M_{m,n}(\mathbb{F})$ and the identity matrix in $M_n(\mathbb{F})$ as $0_{m,n}$ (or 0_n if $m = n$) and I_n, respectively; when the context is clear, we omit the subscripts.

For $A \in M_{m,n}(\mathbb{F})$, denote the *transpose, conjugate,* and *conjugate transpose* of A as $A^\top$, $\overline{A}$, and A^*, respectively. By $\mathrm{Im}(A)$ and $\mathrm{Ker}(A)$, we mean the image of A and kernel of A, respectively. We also denote the *rank* of A, given by the dimension of $\mathrm{Im}(A)$, by $\mathrm{rank}(A)$, and the *nullity* of A, defined as the dimension of $\mathrm{Ker}(A)$, by $\nu(A)$. It is known that $\mathbb{C}^n$ is the direct sum of $\mathrm{Im}(A)$ and $\mathrm{Ker}(A)$ [19, Theorem 2.8.1], and that $n = \mathrm{rank}(A) + \nu(A)$ by the rank-nullity theorem [19, Theorem 2.8.2].

Now, assume that $A \in M_n(\mathbb{C})$. The *spectrum* $\sigma(A)$ of A is the set of eigenvalues, including their multiplicities, of A. If A is Hermitian, denote the jth largest eigenvalue of A as $\lambda_j(A)$ for all $j \in \{1, \ldots, n\}$. That is,

$$\lambda_n(A) \leq \cdots \leq \lambda_1(A).$$

Denote the trace of A as $\mathrm{tr}(A)$ while the determinant of A as $\det(A)$. Note that $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ for any $X \in M_{m,n}(\mathbb{C})$ and $Y \in M_{n,m}(\mathbb{C})$ [20, Subsection 1.1.2], and that $\mathrm{tr}(A) = \sum_{j=1}^{n} \lambda_j$, whenever $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$ [12, Section 1.2]. If $B \in M_m(\mathbb{C})$, we write the *direct sum* of A and B as $A \oplus B \in M_{m+n}(\mathbb{C})$.

2.2 Rank

Recall the following facts about rank.

Proposition 2.1. *[12, Definition 0.4.1] Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Let $A \in M_{m,n}(\mathbb{F})$. The rank of A is the maximum number of linearly independent columns of A.*

Proposition 2.2. *[12, Subsection 0.4.6(a),(d)] Let $A \in M_{m,n}(\mathbb{C})$. Then the following hold:*

1. *$rank(A^*) = rank(A^\top) = rank(\overline{A}) = rank(A)$.*
2. *$rank(A^*A) = rank(A)$.*

Proposition 2.3. *[12, Subsection 0.4.6(b)] Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Let $A \in M_m(\mathbb{F}), B \in M_{m,n}(\mathbb{F})$, and $C \in M_n(\mathbb{F})$. If $rank(A) = m$ and $rank(C) = n$, then*

$$rank(AB) = rank(B) = rank(BC) = rank(ABC).$$

2.3 Norms for Vectors and Matrices

We extend the notion of the usual Euclidean length in $\mathbb{R}^2$ or $\mathbb{R}^3$ by defining a norm on a real or complex vector space.

Definition 2.1. Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. A function $\|\cdot\| : V \to \mathbb{R}$ is a *vector norm* (or simply, a *norm*) if, for all $x, y \in V$ and all $c \in \mathbb{F}$, the following hold:

1. $\|x\| \geq 0$; (nonnegativity)
2. $\|x\| = 0$ if and only if $x = 0$; (positivity)
3. $\|cx\| = |c|\|x\|$; (homogeneity)
4. $\|x + y\| \leq \|x\| + \|y\|$. (triangle inequality)

A vector space V together with a norm $\|\cdot\|$ is called a *normed vector space.* A vector $x \in V$ is a *unit vector* if $\|x\| = 1$.

Example 2.2. Let $A \in M_{m,n}(\mathbb{C})$. Some examples of norms on $M_{m,n}(\mathbb{C})$ are as follows:

1. ℓ_1-norm: $\|A\|_1 = \sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|$

2. Frobenius norm: $\|A\|_2 = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2} = \sqrt{\mathrm{tr}(A^*A)}$.

Note that $\|x\| = \sqrt{x^*x}$ represents the length of a vector $x \in \mathbb{C}^n$ while the angle between two vectors $x, y \in \mathbb{C}^n$ is determined by their usual Euclidean inner product y^*x. The concept of an inner product can be further generalized to any vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ as shown by the following definition.

Definition 2.3. Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. An *inner product* on V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ such that for all $x, y, z \in V$ and all $c \in \mathbb{F}$,

1. $\langle x, x \rangle \geq 0$; (nonnegativity)
2. $\langle x, x \rangle = 0$ if and only if $x = 0$; (positivity)
3. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$; (additivity)
4. $\langle cx, y \rangle = c\langle x, y \rangle$; (homogeneity)
5. $\langle x, y \rangle = \overline{\langle y, x \rangle}$. (Hermitian property)

A vector space V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, together with a given inner product $\langle \cdot, \cdot \rangle$, is called an *inner product space.* Note that the inner product $\langle \cdot, \cdot \rangle$ *induces* a norm $\| \cdot \|$ on V, given by $\|x\| = \sqrt{\langle x, x \rangle}$ for all $x \in V$.

Example 2.4. Some examples of inner products are as follows:

1. Euclidean inner product on $\mathbb{C}^n$: $\langle x, y \rangle = y^*x$ for all $x, y \in \mathbb{C}^n$;
2. Euclidean inner product on $\mathbb{R}^n$: $\langle x, y \rangle = y^\top x$ for all $x, y \in \mathbb{R}^n$;
3. Frobenius inner product on $M_n(\mathbb{C})$: $\langle A, B \rangle = \mathrm{tr}(B^*A)$ for all $A, B \in M_n(\mathbb{C})$.

2.4 Gram-Schmidt Orthogonalization Process

Definition 2.5. Let V be a vector space over a field $\mathbb{K}$. Let $x_1, \ldots, x_k \in V$. The *span* of $x_1, \ldots, x_k$ is the set of all linear combinations of $x_1, \ldots, x_k$ over $\mathbb{K}$, that is, $\mathrm{span}\{x_1, \ldots, x_k\} = \left\{ \sum_{j=1}^{k} \alpha_j x_j : \alpha_j \in \mathbb{K} \text{ for all } j \in \{1, \ldots, k\} \right\}$.

Definition 2.6. Let V be an inner product space. Two vectors $x, y \in V$ are *orthogonal* if $\langle x, y \rangle = 0$. A set of vectors $\{x_1, \ldots, x_k\} \subset \mathbb{C}^n$ is said to be *orthogonal* if $\langle x_i, x_j \rangle = 0$ for all distinct $i, j \in \{1, \ldots, k\}$. In addition, if each x_j is a unit vector, then $\{x_1, \ldots, x_k\}$ is said to be *orthonormal.*

Example 2.7. Let $n \in \mathbb{N}$. For $j \in \{1, \ldots, n\}$, let e_j be the column vector in $\mathbb{C}^n$ whose jth entry is 1 and 0 otherwise. The set $\{e_1, \ldots, e_n\}$ is a basis for $\mathbb{C}^n$ called the *standard basis* for $\mathbb{C}^n$. The standard basis for $\mathbb{C}^n$ is orthonormal since each e_j is a unit vector and $e_j^* e_i = 0$ for each distinct $i, j \in \{1, \ldots, n\}$.

The standard basis for $\mathbb{C}^n$ and its non-trivial subsets are linearly independent. This also applies to any orthonormal set of vectors in $\mathbb{C}^n$.

Proposition 2.4. *[12, Theorem 2.1.2] Every orthonormal set of vectors is linearly independent.*

Note that the converse of Proposition 2.4 does not hold in general. However, by the Gram-Schmidt orthogonalization process, we can replace any linearly independent set $\{v_1, \ldots, v_k\}$ of vectors in an inner product space V by an orthonormal set $\{u_1, \ldots, u_k\}$ of vectors in V with the same span.

Theorem 2.5 (Gram-Schmidt Orthogonalization Process)**.** *[19, Theorem 9.11] Let V be an inner product space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and let $v_1, \ldots, v_k \in V$ be given. Construct $u_1, \ldots, u_k \in V$ in the following manner:*

$$u_1 = v_1,$$
$$u_j = v_j - \sum_{i=1}^{j-1} r_{j,i} u_i,$$

where

$$r_{j,i} = \begin{cases} 0, & \text{if } u_i = 0, \\ \frac{\langle v_j, u_i \rangle}{\langle u_i, u_i \rangle}, & \text{if } u_i \neq 0, \end{cases}$$

for all $j \in \{1, \ldots, k\}$. Then $\{u_1, \ldots, u_k\}$ is orthogonal with the property that for all $j \in \{1, \ldots, k\}$,

$$\text{span}\{u_1, \ldots, u_j\} = \text{span}\{v_1, \ldots, v_j\}.$$

Also, for all $j \in \{1, \ldots, k\}$, $u_j = 0$ if and only if $v_j \in \text{span}\{v_1, \ldots, v_{j-1}\}$.

2.5 Normal Matrices

The following are special types of matrices relevant to the study.

Definition 2.8. Let $A = [a_{ij}] \in M_n(\mathbb{C})$. Then A is said to be a *diagonal matrix* if $a_{ij} = 0$ for all distinct $i, j \in \{1, \ldots, n\}$. In this case, we use the notation $A = \text{diag}(a_{11}, \ldots, a_{nn})$.

Observe that if $A \in M_n(\mathbb{C})$ is a diagonal, then $A^* = \overline{A}$ and that $A\overline{A} = \overline{A}A$. This motivates the following definition.

Definition 2.9. Let $A \in M_n(\mathbb{C})$. Then A is said to be

- *normal* if A commutes with its conjugate transpose, that is, $AA^* = A^*A$,
- *unitary* if $A^*A = I_n$,
- *Hermitian* if $A^* = A$, and
- *skew-Hermitian* if $A^* = -A$.

It follows from the definition that unitary, Hermitian, and skew-Hermitian matrices are normal.

Definition 2.10. Let $A, B \in M_n(\mathbb{C})$. Then A, B are said to be *unitarily similar* if $B = UAU^*$ for some unitary $U \in M_n(\mathbb{C})$.

Observe that if $A, B \in M_n(\mathbb{C})$ are unitarily similar, then $\mathrm{tr}(A) = \mathrm{tr}(B)$ and A is Hermitian if and only if B is Hermitian. The same is true if 'Hermitian' is replaced by skew-Hermitian, unitary, or normal.

2.6 Projections

Definition 2.11. Let $P \in M_n(\mathbb{C})$. Then P is a (Hermitian) projection if P is both Hermitian and idempotent, that is, $P^2 = P = P^*$.

The following are some observations on projections.

Proposition 2.6. *Let $P \in M_n(\mathbb{C})$ be a projection of rank k. Then the following hold:*

1. *$\sigma(P) \subseteq \{0, 1\}$.*
2. *$Py = y$ for all $y \in Im(P)$.*
3. *If $u \in \mathbb{C}^n$ is a column of P, then $Pu = u$.*
4. *If $Q \in M_n(\mathbb{C})$ is unitarily similar to P, then Q is a projection of rank k.*

Proof.

1. Let $0 \neq x \in \mathbb{C}^n$ be an eigenvector of P corresponding to an eigenvalue λ of P. Then $\lambda x = Px = P^2x = P(\lambda x) = \lambda(Px) = \lambda^2 x$. Since x is nonzero, $\lambda \in \{0, 1\}$.

2. Let $y \in \mathrm{Im}(P)$. Then there exists $x \in \mathbb{C}^n$ such that $Px = y$. Thus, $Py = P(Px) = P^2x = Px = y$.

3. Suppose $u \in \mathbb{C}^n$ is a column of P. By the second part, $Pu = u$ since $u = Pe_j \in \mathrm{Im}(P)$ for some $j \in \{1, \ldots, n\}$.

4. Suppose $Q = UPU^*$ for some unitary matrix $U \in M_n(\mathbb{C})$. Then,

$$Q^2 = (UPU^*)^2 = UP^2U^* = UPU^* = Q,$$

and

$$Q^* = (UPU^*)^* = UP^*U^* = UPU^* = Q.$$

Hence, Q is a projection and rank Q = rank $P = k$.

□

The following is a matrix decomposition theorem for projections.

Proposition 2.7. *Let $P \in M_n(\mathbb{C})$. If P is a projection of rank k, then there exists $V \in M_{n,k}(\mathbb{C})$ with k orthonormal columns and unitary $U \in M_n(\mathbb{C})$ such that $P = VV^* = U(I_k \oplus 0_{n-k})U^*$.*

Proof. Suppose P is a projection of rank k. Then there exists an orthonormal basis $\{x_j\}_{j=1}^k$ for $\mathrm{Im}(P)$ such that $Px_j = x_j$ for all $j \in \{1, \ldots, k\}$. Moreover, if we set $V = [x_1 \ \cdots \ x_k] \in M_{n,k}(\mathbb{C})$, we have $V^*V = I_k$ and so

$$PV = P[x_1 \ \cdots \ x_k] = [Px_1 \ \cdots \ Px_k] = [x_1 \ \cdots \ x_k] = V.$$

If $k = n$, then V is unitary, hence normal, and so $P = PVV^* = VV^*$. We can choose $U = V$ so that $P = VV^* = UI_nU^*$.

If $k < n$, then we extend $\{x_j\}_{j=1}^k$ to an orthonormal basis $\{x_j\}_{j=1}^n$ for $\mathbb{C}^n$ to form a unitary matrix $U = [V \ \ W] \in M_n(\mathbb{C})$ where $W = [x_{k+1} \ \cdots \ x_n]$. Since $\mathbb{C}^n$ is the direct sum of $\mathrm{Im}(P)$ and $\mathrm{Ker}(P)$, and $\{x_j\}_{j=k+1}^n$ is a basis for $\mathrm{Ker}(P)$,

we get $PW = 0_{n,n-k}$. Furthermore,

$$\begin{aligned} U^*PU &= \begin{bmatrix} V^* \\ W^* \end{bmatrix} P[V \quad W] \\ &= \begin{bmatrix} V^*(PV) & V^*(PW) \\ (W^*P)V & W^*(PW) \end{bmatrix} \\ &= \begin{bmatrix} V^*V & V^*0_{n,n-k} \\ 0_{n-k,n}V & W^*0_{n,n-k} \end{bmatrix} \\ &= I_k \oplus 0_{n-k}. \end{aligned}$$

Hence, $P = U(I_k \oplus 0_{n-k})U^* = [V \quad W](I_k \oplus 0_{n-k}) \begin{bmatrix} V^* \\ W^* \end{bmatrix} = VV^*$.

□

Now, suppose $V \in M_{n,k}(\mathbb{C})$ has k orthonormal columns and $U \in M_n(\mathbb{C})$ is a unitary matrix. Then $V^*V = I_k$ and so

$$(VV^*)^2 = (VV^*)(VV^*) = V(V^*V)V^* = VI_kV^* = VV^*$$

and

$$\operatorname{rank} VV^* = \operatorname{rank} V^* = \operatorname{rank} V = k.$$

Since VV^* is Hermitian, this implies that VV^* is a projection of rank k. Likewise, $U(I_k \oplus 0_{n-k})U^*$ is a projection of rank k since $I_k \oplus 0_{n-k}$ is a projection of rank k.

As a consequence of Proposition 2.7, we obtain some fundamental properties of projections in addition to Proposition 2.6.

Proposition 2.8. *Let $P, Q \in M_n(\mathbb{C})$ be projections. The following hold:*

1. *If P and Q have the same rank, then P and Q are unitarily similar.*
2. *If P has rank n, then $P = I_n$.*
3. *$tr(P) = rank(P)$.*

Proof. Let $k = \operatorname{rank}(P)$. For the first statement, suppose $\operatorname{rank}(Q) = k$. By Proposition 2.7, both P and Q are unitarily similar to $I_k \oplus 0_{n-k}$. Since unitary similarity is an equivalence relation on $M_n(\mathbb{C})$ [12, Section 0.11], this implies that P and Q are unitarily similar.

For the second part, suppose $k = n$. Since I_n is also a projection of rank n, the first statement implies that $P = U^*I_nU = I_n$ for some unitary $U \in M_n(\mathbb{C})$.

For the last part, $\operatorname{tr}(P) = \operatorname{tr}(I_k \oplus 0_{n-k}) = k = \operatorname{rank}(P)$ since P and $I_k \oplus 0_{n-k}$ are unitarily similar by Proposition 2.7. □

2.7 Decomposition of Matrices

The following are some matrix decomposition theorems used in this study.

Theorem 2.9 (Spectral Theorem for Normal Matrices)**.** *[19, Theorem 10.9] Let $A \in M_n(\mathbb{C})$ and $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$. The following are equivalent:*

1. *A is normal.*
2. *A is unitarily diagonalizable, that is, there exists a unitary $U \in M_n(\mathbb{C})$ and a diagonal $D = diag(\lambda_1, \ldots, \lambda_n) \in M_n(\mathbb{C})$ such that $A = UDU^*$.*
3. *A has orthogonal spectral resolution, that is, $A = \sum_{j=1}^{n} \lambda_j P_j$ for some projection P_j of rank 1 for all $j \in \{1, \ldots, n\}$, with $P_i P_j = 0_n$ whenever $i \neq j$, and $\sum_{j=1}^{n} P_j = I_n$.*

A representation of a normal matrix $A \in M_n(\mathbb{C})$ in the second part of the above theorem is called a *spectral decomposition* of A. As a consequence of Theorem 2.9, we obtain a more general result.

Theorem 2.10 (Singular Value Decomposition)**.** *[12, Theorem 2.6.3] Let $A \in M_{m,n}(\mathbb{C})$ be given, let $q = \min\{m, n\}$, and suppose rank $A = r > 0$. There are unitary matrices $V \in M_m(\mathbb{C})$ and $W \in M_n(\mathbb{C})$, and a square diagonal matrix*

$$\Sigma_q = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_q \end{bmatrix} \in M_q(\mathbb{C})$$

such that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_q$ and $A = V\Sigma W^*$, in which

$$\Sigma = \Sigma_q \text{ if } m = n,$$
$$\Sigma = \begin{bmatrix} \Sigma_q & 0 \end{bmatrix} \in M_{m,n}(\mathbb{C}) \text{ if } m < n,$$
$$\Sigma = \begin{bmatrix} \Sigma_q \\ 0 \end{bmatrix} \in M_{m,n}(\mathbb{C}) \text{ if } m > n.$$

The diagonal entries of Σ_q in Theorem 2.10 are the *singular values* of A. Their squares are the eigenvalues of A^*A or AA^*.

Theorem 2.11 (Toeplitz Decomposition)**.** *[12, Theorem 4.1.2] Let $A \in M_n(\mathbb{C})$. Then A can be written uniquely as $A = H + iK$ where $H \in M_n(\mathbb{C})$ and $K \in M_n(\mathbb{C})$ are both Hermitian.*

The above decomposition theorem motivates the following definition.

Definition 2.12. Let $A \in M_n(\mathbb{C})$. The *real part of A* as $\mathfrak{Re}(A) = \frac{1}{2}(A + A^*)$ and the *imaginary part of A* as $\mathfrak{Im}(A) = \frac{1}{2i}(A - A^*)$.

For any $A, B \in M_n(\mathbb{C})$, observe that

1. $\mathfrak{Re}(A + B) = \mathfrak{Re}(A) + \mathfrak{Re}(B)$ and
2. $x^*\mathfrak{Re}(A)x = \mathfrak{Re}(x^*Ax)$ for all $x \in \mathbb{C}^n$.
3. In particular, given $\{x_1, \ldots, x_k\} \subset \mathbb{C}^n$, we have
$$\mathfrak{Re}\left(\sum_{j=1}^{k} x_j^* A x_j\right) = \sum_{j=1}^{k} x_j^* \mathfrak{Re}(A) x_j.$$

2.8 Convergence in a Normed Vector Space

In a normed vector space, we have the following notion of convergence.

Definition 2.13. Let V be a normed vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. A sequence $\{x_j\}_{j=1}^{\infty}$ of vectors in V *converges* to a vector $x \in V$ if $\lim_{j\to\infty} \|x_j - x\| = 0$. If $\{x_j\}_{j=1}^{\infty}$ converges to x with respect to $\|\cdot\|$, we write $\lim_{j\to\infty} x_j = x$ (or $x_j \to x$) with respect to $\|\cdot\|$.

It is known that the limit of a sequence with respect to a given norm is unique if it exists.

Proposition 2.12. *[12, Corollary 5.4.6] If $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are norms on a finite-dimensional real or complex vector space V, and if $\{x_j\}_{j=1}^{\infty}$ is a given sequence of vectors in V, then $\lim_{j\to\infty} x_j = x$ with respect to $\|\cdot\|_\alpha$ if and only if $\lim_{j\to\infty} x_j = x$ with respect to $\|\cdot\|_\beta$.*

In other words, all norms in a finite-dimensional real or complex vector space are *equivalent.* As a consequence, the convergence of a sequence of matrices in $M_{m,n}(\mathbb{C})$ is equivalent to the convergence of its entries.

Proposition 2.13. *Let $\{A^{(\ell)}\}_{\ell=1}^{\infty}$ be a sequence of matrices in $M_{m,n}(\mathbb{C})$ where $A^{(\ell)} = [a_{ij}^{(\ell)}]$ for all $\ell \in \mathbb{N}$. Then $\lim_{\ell\to\infty} A^{(\ell)} = A = [a_{ij}]$ if and only if $\lim_{\ell\to\infty} a_{ij}^{(\ell)} = a_{ij}$ for each $i \in \{1, \ldots, m\}$ and each $j \in \{1, \ldots, n\}$.*

Proof. Suppose $\lim_{\ell\to\infty} A^{(\ell)} = A = [a_{ij}]$. Note that $M_{m,n}(\mathbb{F})$ is a finite-dimensional vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. By Proposition 2.12, we assume, without loss of generality, that $\lim_{\ell\to\infty} A^{(\ell)} = A$ with respect to $\|\cdot\|_1$, that is, $\|A^{(\ell)} - A\|_1 \to 0$. In particular, for any $\varepsilon > 0$, there exists $L > 0$ such that whenever $\ell \geq L$,

$$|a_{ij}^{(\ell)} - a_{ij}| \leq \sum_{r=1}^{n}\sum_{s=1}^{n} |a_{rs}^{(\ell)} - a_{rs}| < \varepsilon$$

for all $i \in \{1,\ldots,m\}$ and all $j \in \{1,\ldots,n\}$. Thus, $\lim_{\ell\to\infty} a_{ij}^{(\ell)} = a_{ij}$ for each $i \in \{1,\ldots,m\}$ and each $j \in \{1,\ldots,n\}$.

Conversely, suppose $\lim_{\ell\to\infty} a_{ij}^{(\ell)} = a_{ij}$ for all $i \in \{1,\ldots,m\}$ and all $j \in \{1,\ldots,n\}$. Then for each i, j and all $\varepsilon > 0$, there exists $L_{ij} > 0$ such that $|a_{ij}^{(\ell)} - a_{ij}| < \dfrac{\varepsilon}{n^2}$ whenever $\ell \geq L_{ij}$. Choose $L = L_{pq}$ where

$$|a_{pq}^{(\ell)} - a_{pq}| = \max\{|a_{ij}^{(\ell)} - a_{ij}| : i \in \{1,\ldots,m\} \text{ and } j \in \{1,\ldots,n\}\}$$

so that when $\ell \geq L$, we have

$$\|A^{(\ell)} - A\|_1 = \sum_{i=1}^{n}\sum_{j=1}^{n} |a_{ij}^{(\ell)} - a_{ij}| < n^2\left(\frac{\varepsilon}{n^2}\right) = \varepsilon,$$

that is, $A^{(\ell)} \to A$ in $M_{m,n}(\mathbb{C})$. □

As a consequence, we have the following corollary.

Corollary 2.14. *Let $\{A^{(\ell)}\}_{\ell=1}^{\infty}$ and $\{B^{(\ell)}\}_{\ell=1}^{\infty}$ be sequences of matrices in $M_{m,n}(\mathbb{C})$ where $A^{(\ell)} = [a_{ij}^{(\ell)}]$ and $B^{(\ell)} = [b_{ij}^{(\ell)}]$ for all $\ell \in \mathbb{N}$. If $\lim_{\ell\to\infty} A^{(\ell)} = A = [a_{ij}]$ and $\lim_{\ell\to\infty} B^{(\ell)} = B = [b_{ij}]$, the following hold:*

1. $\lim_{\ell\to\infty} (A^{(\ell)})^* = A^*$;
2. $\lim_{\ell\to\infty} A^{(\ell)}B^{(\ell)} = AB$.

Proof. Suppose $\lim_{\ell\to\infty} A^{(\ell)} = A = [a_{ij}]$ and $\lim_{\ell\to\infty} B^{(\ell)} = B = [b_{ij}]$.

For the first part, note that $a_{ij}^{(\ell)} \to a_{ij}$ for each $i \in \{1,\ldots,m\}$ and each $j \in \{1,\ldots,n\}$ by Proposition 2.13. By definition, for all $\varepsilon > 0$, there exists $L > 0$ such that whenever $\ell \geq L$, $|\overline{a_{ji}^{(\ell)}} - \overline{a_{ji}}| = |\overline{a_{ji}^{(\ell)} - a_{ji}}| = |a_{ji}^{(\ell)} - a_{ji}| < \varepsilon$. By Proposition 2.13, $\lim_{\ell\to\infty} (A^{(\ell)})^* = A^*$.

For the second part, note that

$$\lim_{\ell\to\infty} \underbrace{\sum_{r=1}^{n} a_{ir}^{(\ell)} b_{rj}^{(\ell)}}_{(i,j) \text{ entry of } A^{(\ell)}B^{(\ell)}} = \sum_{r=1}^{n} \lim_{\ell\to\infty} a_{ir}^{(\ell)} b_{rj}^{(\ell)} = \underbrace{\sum_{r=1}^{n} a_{ir}b_{rj}}_{(i,j) \text{ entry of } AB}$$

for all $i \in \{1, \ldots, m\}$ and all $j \in \{1, \ldots, n\}$. Thus, $\lim_{\ell \to \infty} A^{(\ell)} B^{(\ell)} = AB$ by Proposition 2.13.

□

2.9 Continuous Functions on a Normed Vector Space

The notion of continuity in a normed vector space is given below.

Definition 2.14. Let $f : X \to Y$ be a function such that X and Y are real or complex vector spaces with given norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$, respectively. The function f is continuous at a point $x_0 \in X$ if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that $\|f(x) - f(x_0)\|_\beta < \varepsilon$ whenever $x \in X$ and $0 \leq \|x - x_0\|_\alpha < \delta$. Moreover, f is continuous on X if f is continuous at every point $x_0 \in X$.

Note that if X is a real or complex vector space with a given norm $\|\cdot\|$, then the function

$$\begin{aligned} d : X \times X &\to \mathbb{R} \\ (x, y) &\mapsto \|x - y\| \end{aligned}$$

is a metric on X. Hence, X is metrizable. In particular, we have the following result.

Theorem 2.15 (Sequential Criterion of Continuity)**.** *[18, Theorem 21.3] Let $f : X \to Y$ be a function such that X and Y are real or complex vector spaces with given norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$, respectively. Then f is continuous on X if and only if for every convergent sequence $x_j \to x$ in X, the sequence $f(x_j) \to f(x)$ in Y.*

As a consequence, we have the following corollary.

Corollary 2.16. *Let $A \in M_n(\mathbb{C})$. The function $f : \mathbb{C}^n \to \mathbb{C}$ defined by $x \mapsto x^* A x$ is continuous.*

Proof. Let $\{x_j\}_{j=1}^\infty$ be a sequence of vectors in $\mathbb{C}^n$ such that $x_j \to x$ for some $x \in \mathbb{C}^n$. Let $y_j = Ax_j$ for all $j \in \mathbb{N}$. Let $y = Ax$. By Corollary 2.14, $y_j \to y$ and $x_j^* \to x^*$. Thus, $f(x_j) = x_j^* A x_j = x_j^* y_j \to x^* y = x^* A x = f(x)$. By Theorem 2.15, f is continuous. □

2.10 Compact Sets

Definition 2.15. Let V be a normed vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. A set $S \subseteq V$ is

- *open* if, for each $x \in S$, there exists $\delta > 0$ such that

$$\{y \in V : \|y - x\| < \delta\} \subseteq S,$$

- *closed* if the complement of S in V is open,
- *bounded* if there exists $r > 0$ such that $\|x\| \leq r$ for all $x \in S$, and
- *compact* if every sequence in S has a subsequence that converges to a vector $x \in S$.

Equivalently, a set $S \subseteq V$ is closed if and only if for any sequence of points $\{x_n\}$ in S, $x_n \to x$ implies $x \in S$ [12, Appendix E]. In addition, by the Heine-Borel theorem, a set $S \subseteq \mathbb{R}^n$ is compact if and only if S is both closed and bounded [18, Theorem 27.3].

Proposition 2.17. *[18, Theorem 26.5] The image of a compact space under a continuous map is compact.*

2.11 Convex Sets

Definition 2.16. Let V be a real vector space. A set $K \subseteq V$ is *convex* if, for any given pair of points $x, y \in K$, the set $\{(1-t)\,x + yt : t \in [0,1]\} \subseteq K$.

Example 2.17. Below are some examples of convex sets.

1. The empty set $\varnothing$ is convex since the condition for convexity is satisfied vacuously.
2. Given $x, y \in V$, the sets $\{tx + (1-t)y : t \in [0,1]\}$ and $\{tx + (1-t)y : t \in \mathbb{R}\}$ are convex.
3. Closed half-planes in $\mathbb{C}$, which are sets of the form $\{z \in \mathbb{C} : \mathfrak{Re}(e^{i\theta} z) \leq a\}$ for some $\theta \in [0, 2\pi)$ and $a \in \mathbb{R}$, are convex. Indeed, if $x, y \in \mathbb{C}$, then for all

$t \in [0,1]$, we have

$$\begin{aligned}\mathfrak{Re}(e^{i\theta}(tx+(1-t)y)) &= \mathfrak{Re}(e^{i\theta}tx) + \mathfrak{Re}(e^{i\theta}(1-t)y) \\ &= t\mathfrak{Re}(e^{i\theta}x) + (1-t)\mathfrak{Re}(e^{i\theta}y) \\ &\leq ta + (1-t)a \\ &= a.\end{aligned}$$

The union of two convex sets may not be convex in general. However, the following proposition says that an arbitrary intersection of convex sets is always convex.

Proposition 2.18. *Let V be a real vector space. Let J be an index set and let $K_\alpha \subseteq V$ be convex for all $\alpha \in J$. Then $\bigcap_{\alpha\in J} K_\alpha$ is convex.*

Proof. Let $x, y \in \bigcap_{\alpha\in J} K_\alpha$, that is, $x, y \in K_\alpha$ for all $\alpha \in J$. Let $t \in [0,1]$. By the convexity of each K_α, $tx+(1-t)y \in K_\alpha$ for all $\alpha \in J$, that is, $tx+(1-t)y \in \bigcap_{\alpha\in J} K_\alpha$. Thus, $\bigcap_{\alpha\in J} K_\alpha$ is convex. □

A specific type of linear combination is associated with convex sets. These are called convex combinations and are used to characterize convex sets.

Definition 2.18. Let V be a real vector space. Let $x_1, \ldots, x_n \in V$. A *convex combination* of $x_1, \ldots, x_n$ is of the form $\sum_{j=1}^{n} \alpha_j x_j$ where $\alpha_j \geq 0$ for all $j \in \{1, \ldots, n\}$ and $\sum_{j=1}^{n} \alpha_j = 1$.

Proposition 2.19. *Let V be a real vector space. A set $K \subseteq V$ is convex if and only if any convex combination of a finite list of vectors in K lies in K.*

Proof. Let $\sum_{j=1}^{n} \alpha_j x_j$ be a convex combination of $x_1, \ldots, x_n \in K$, that is, $\alpha_j \geq 0$ for all $j \in \{1, \ldots, n\}$ and $\sum_{j=1}^{n} \alpha_j = 1$.

To show the forward implication, we proceed by induction.

Basis Step. Suppose $n = 1$. Then $\alpha_1 = 1$. In particular, $\alpha_1 x_1 = x_1 \in K$.

Inductive Step. Suppose the claim holds for $n = k$. We want to show that the claim holds for $n = k + 1$.

Indeed, suppose $\alpha_{k+1} < 1$. Since $\alpha_j \geq 0$ for all $j \in \{1, \ldots, k+1\}$, then $\dfrac{\alpha_j}{1-\alpha_{k+1}} \geq 0$ for all $j \in \{1, \ldots, k\}$ and $\displaystyle\sum_{j=1}^{k} \frac{\alpha_j}{1-\alpha_{k+1}} = 1$. By inductive hypothesis, $\displaystyle\sum_{j=1}^{k} \frac{\alpha_j}{1-\alpha_{k+1}} x_j \in K$. By the convexity of K, we have

$$\sum_{j=1}^{k+1} \alpha_j x_j = (1-\alpha_{k+1}) \sum_{j=1}^{k} \frac{\alpha_j}{1-\alpha_{k+1}} x_j + \alpha_{k+1} x_{k+1} \in K.$$

Now, suppose $\alpha_{k+1} = 1$. Then $\alpha_j = 0$ for $j \in \{1, \ldots, k\}$. In particular,

$$\sum_{j=1}^{k+1} \alpha_j x_j = \sum_{j=1}^{k} \alpha_j x_j + \alpha_{k+1} x_{k+1} = 0 + x_{k+1} = x_{k+1} \in K.$$

This proves the inductive hypothesis. Therefore, $\displaystyle\sum_{j=1}^{n} \alpha_j x_j \in K$ for all $n \in \mathbb{N}$.

Conversely, suppose $\displaystyle\sum_{j=1}^{n} \alpha_j x_j \in K$ for all $n \in \mathbb{N}$. If $n = 2$, then $\alpha_2 = 1 - \alpha_1$. Thus, $\alpha_1 x_1 + \alpha_2 x_2 = \alpha_1 x_1 + (1-\alpha_1) x_2 \in K$. Hence, K is convex. □

Definition 2.19. Let V be a real vector space. Let $K \subseteq V$ be a convex set. An *extreme point* of K is a point $z \in K$ such that $z = tx + (1-t)y, t \in (0,1), x, y \in K$ implies $x = y = z$. Denote the set of all extreme points of K by $\mathrm{Ext}(K)$.

The following gives a characterization of an extreme point.

Proposition 2.20. *[13, Theorem 3.7.6] Let V be a real vector space. Let $K \subseteq V$ be a nonempty compact convex set. Then $x_0 \in K$ is an extreme point of K if and only if $K \setminus \{x_0\}$ is convex.*

Example 2.20. Let V be a real vector space. Let $x, y \in V$ be distinct.

1. If $K = \{tx + (1-t)y : t \in [0,1]\}$, then $\mathrm{Ext}(K) = \{x, y\}$ since $K \setminus \{x\}$ and $K \setminus \{y\}$ are convex while $K \setminus \{z\}$ is not convex for $z \notin \{x, y\}$.

2. If $K = \{tx + (1-t)y : t \in \mathbb{R}\}$, then $\mathrm{Ext}(K) = \varnothing$ since for all $t \in \mathbb{R}$,

$$tx + (1-t)y = \frac{1}{2}[\underbrace{(t+1)x - ty}_{\in K}] + \frac{1}{2}[\underbrace{(t-1)x + (2-t)y}_{\in K}].$$

Note that $\mathrm{Ext}(K)$ is not always nonempty. This may happen if K is a closed half-plane or K is an open disk. The following proposition gives a sufficient condition for the existence of an extreme point.

Proposition 2.21. *[13, Theorem 3.7.11] Let V be a finite-dimensional real vector space. If K is a nonempty compact convex subset of V, then K has at least one extreme point.*

Definition 2.21. Let V be a real vector space. Let $S \subseteq V$. The *convex hull* of S is the intersection of all convex sets containing S. The convex hull is denoted by $\operatorname{co}(S)$.

The convex hull $\operatorname{co}(S)$ of any subset S of a real vector space V is the collection of all convex combinations of points in S. Note that $\operatorname{co}(S)$ is always convex by Proposition 2.18 and is the smallest convex set to contain S.

Example 2.22. Let $S \subseteq \mathbb{C}$.

1. If $S = \{a, b\}$ where $a, b \in \mathbb{C}$ are distinct, then the convex hull of S is the line segment joining a and b.

2. If S is a finite set of collinear points in $\mathbb{C}$ or $\mathbb{R}$, then the convex hull of S is the line segment joining a pair of points in S with the largest distance.

3. If $S = \{\pm 1, 2i\}$, then the convex hull of S is the isosceles triangle (including its interior) having the points of S as its vertices (see Figure 2.1).

4. If $S = \{0, 1 \pm i, -1 \pm i\}$, then the convex hull of S is the square (including its interior) having the points of $S \setminus \{0\}$ as its vertices (see Figure 2.2).

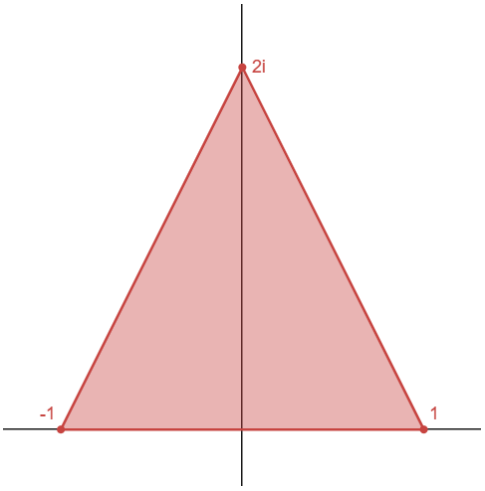


Figure 2.1: Convex hull of $S = \{\pm 1, 2i\}$

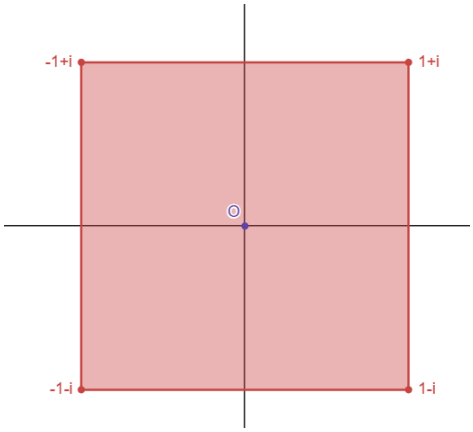


Figure 2.2: Convex hull of $S = \{0, 1 \pm i, -1 \pm i\}$

Theorem 2.22 (Krein-Milman)**.** *[20, Theorem* 1.10*] Let V be a finite-dimensional real vector space. Let $K \subseteq V$ be a nonempty compact convex set. Then $K = co(Ext(K))$.*

In particular, every point in K can be expressed as a convex combination of its extreme points. Thus, the *generators* of K are its extreme points.

Theorem 2.23 (Separating Hyperplane Theorem on $\mathbb{R}^n$)**.** *[2, Section 2.5.1] Let K and L be nonempty disjoint convex sets of $\mathbb{R}^n$. Then there exist nonzero $y \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that $y^\top x \leq b$ for all $x \in K$ and $y^\top x \geq b$ for all $x \in L$.*

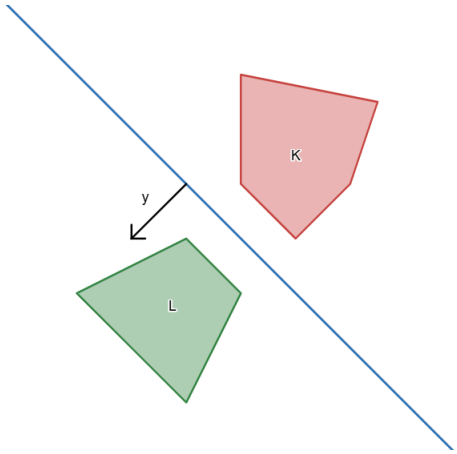


Figure 2.3: Two convex sets K and L in $\mathbb{R}^2$ separated by a hyperplane determined by y

Since singleton sets are convex, we have the following corollary.

Corollary 2.24. *[2, Example 2.20] Let K be a closed convex subset of $\mathbb{R}^n$ and $x_0 \notin K$. There exist nonzero $y \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that $y^\top x < b$ for all $x \in K$ and $y^\top x_0 > b$.*

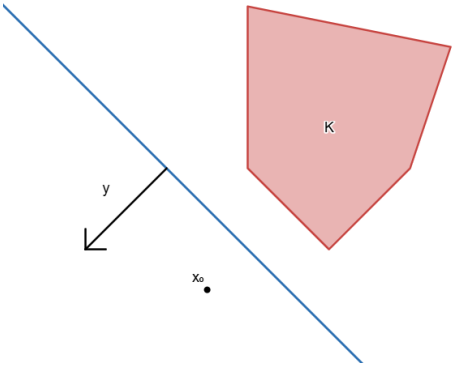


Figure 2.4: A convex set K and a point x_0 in $\mathbb{R}^2$ separated by a hyperplane determined by y

Without loss of generality, we can assume in Corollary 2.24 that $y \in \mathbb{R}^n$ is a unit vector since y is nonzero. Thus, if $n = 2$, we can assume that y is a point in the unit circle, that is, $y = [\cos(-\varphi) \quad \sin(-\varphi)]^\top = [\cos\varphi \quad -\sin\varphi]^\top \in \mathbb{R}^2$ for some $\varphi \in [0, 2\pi)$.

2.12 Variational Characterization of the Eigenvalues of a Hermitian matrix

The following are some important properties of Hermitian matrices.

Proposition 2.25. *[12, Theorem 4.1.3(a)-(b)] Let $A \in M_n(\mathbb{C})$ be Hermitian. The following hold:*

1. *$x^*Ax \in \mathbb{R}$ for all $x \in \mathbb{C}^n$.*
2. *The eigenvalues of A are all real.*

Theorem 2.26 (Rayleigh)**.** *[12, Theorem 4.2.2] Let $A \in M_n(\mathbb{C})$ be Hermitian. Then for any unit vector $x \in \mathbb{C}^n$, $\lambda_n(A) \leq x^*Ax \leq \lambda_1(A)$, with equality on the right hand (respectively, left hand) if and only if $Ax = \lambda_1(A)x$ (respectively, $Ax = \lambda_n(A)x$). Moreover, $\lambda_1(A) = \max\{x^*Ax : x \in \mathbb{C}^n$ is a unit vector$\}$ and $\lambda_n(A) = \min\{x^*Ax : x \in \mathbb{C}^n$ is a unit vector$\}$.*

2.13 Rearrangement Inequality of Real Numbers

For each $n \in \mathbb{N}$, the symmetric group on n letters is the set of all bijections on $\{1, \ldots, n\}$.

Proposition 2.27. *[10, Theorem 368] Let $a_n \leq \cdots \leq a_2 \leq a_1$ and $b_n \leq \cdots \leq b_2 \leq b_1$ be real numbers. Then*

$$\sum_{j=1}^{n} a_{n-j+1} b_j \leq \sum_{j=1}^{n} a_{\tau(j)} b_j \leq \sum_{j=1}^{n} a_j b_j$$

for all permutation $\tau \in S_n$.

As a consequence, we obtain the following corollary.

Corollary 2.28. *Let $a_n \leq \cdots \leq a_2 \leq a_1$ be real numbers. Then for all $k \in \{1, \ldots, n\}$,*

$$\sum_{j=1}^{k} a_{n-k+j} \leq \sum_{j=1}^{k} a_{\tau(j)} \leq \sum_{j=1}^{k} a_j$$

for all permutation $\tau \in S_n$,

Proof. For $k \in \{1, \ldots, n\}$, define $b_j = 1$ for all $j \in \{1, \ldots, k\}$ and $b_j = 0$ for all $j \in \{k+1, \ldots, n\}$. By Proposition 2.27, $\sum_{j=1}^{k} a_{\tau(j)} \leq \sum_{j=1}^{k} a_j$. Since $-a_1 \leq \cdots \leq -a_n$,

$$-\sum_{j=1}^{k} a_{\tau(j)} = \sum_{j=1}^{k} (-a_{\tau(j)}) \leq \sum_{j=1}^{k} (-a_{n-k+j}) = -\sum_{j=1}^{k} a_{n-k+j},$$

that is, $\sum_{j=1}^{k} a_{n-k+j} \leq \sum_{j=1}^{k} a_{\tau(j)}$ for all $\tau \in S_n$. Combining the two inequalities establishes the claim. □

Chapter 3

The Classical Numerical Range

In this chapter, we introduce the notion of the numerical range: a set of complex numbers associated with a given matrix $A \in M_n(\mathbb{C})$. The numerical range provides more information about A that the spectrum alone cannot give.

Definition 3.1. Let $A \in M_n(\mathbb{C})$. The *numerical range* of A is the set

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n \text{ is a unit vector}\}.$$

To discuss the properties of the numerical range, we have the following definition.

Definition 3.2. Given any two subsets R and S of $\mathbb{C}$, we define the *sum of* R and S as the set $R + S = \{r + s : r \in R \text{ and } s \in S\}$. If $R = \{r\}$, we write $r + S := R + S$. For any $z \in \mathbb{C}$, we define the *z-multiple of* R as the set $zR = \{zr : r \in R\}$. Whenever $z = -1$, we write $-R := -1 \cdot R$. In addition, we define the *real part of* S as $\mathfrak{Re}(S) = \{\mathfrak{Re}(s) : s \in S\}$.

The following proposition lists some of the properties of the numerical range.

Proposition 3.1. *[11, Section 1.2] Let $A \in M_n(\mathbb{C})$. The following hold:*

1. $W(\alpha I_n + A) = \alpha + W(A)$ *for any* $\alpha \in \mathbb{C}$; (*translation*)
2. $W(\alpha A) = \alpha W(A)$ *for any* $\alpha \in \mathbb{C}$; (*scalar multiplication*)
3. $W(\mathfrak{Re}(A)) = \mathfrak{Re}(W(A))$; (*projection*)
4. $\sigma(A) \subseteq W(A)$; (*spectral containment*)
5. $W(U^*AU) = W(A)$ *for any unitary* $U \in M_n(\mathbb{C})$. (*unitary invariance*)

Proof.

For the first part, we have

$$\begin{aligned} W(\alpha I_n + A) &= \{x^*(\alpha I_n + A)x : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \{\alpha x^*x + x^*Ax : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \{\alpha + x^*Ax : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \alpha + W(A). \end{aligned}$$

For the second part, we have

$$\begin{aligned} W(\alpha A) &= \{x^*(\alpha A)x : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \{\alpha x^* Ax : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \alpha W(A). \end{aligned}$$

For the third part, we have

$$\begin{aligned} W(\mathfrak{Re}(A)) &= \{x^*\mathfrak{Re}(A)x : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \{\mathfrak{Re}(x^*Ax) : x \in \mathbb{C}^n \text{ is a unit vector}\} \\ &= \mathfrak{Re}(W(A)). \end{aligned}$$

For the fourth part, suppose $\lambda \in \sigma(A)$. There exists unit vector $x \in \mathbb{C}^n$ such that $Ax = \lambda x$. Hence, $\lambda = \lambda x^* x = x^*(\lambda x) = x^* Ax \in W(A)$ and so $\sigma(A) \subseteq W(A)$.

For the last part, if $x \in \mathbb{C}^n$ is a unit vector, then $x^*(U^*AU)x = y^*Ay \in W(A)$, where $y = Ux$, so $y^*y = x^*U^*Ux = x^*x = 1$. Thus, $W(U^*AU) \subseteq W(A)$. In particular, since U^* is unitary, $W(A) = W(U(U^*AU)U^*) \subseteq W(U^*AU)$. Hence, $W(A) = W(U^*AU)$. □

To illustrate Proposition 3.1, we provide the following examples.

Example 3.3. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. On the figure below, $W(\frac{1}{2}A)$ is obtained by shrinking $W(A)$ (the larger red triangular region) by a factor of $\frac{1}{2}$.

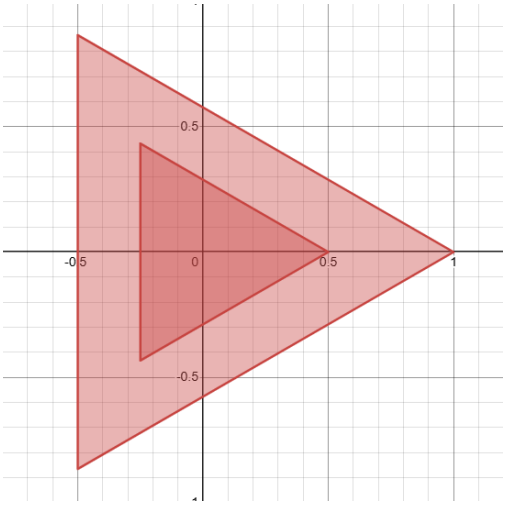


Figure 3.1: $W(A)$ vs. $W(\frac{1}{2}A)$

Example 3.4. Let $B = 2iI_3 + A$ and consider the figure below. Notice that $W(B)$ is formed by translating $W(A)$ two units upwards along the imaginary axis, while $W(\mathfrak{Re}(B))$ is formed by projecting $W(B)$ to the real axis.

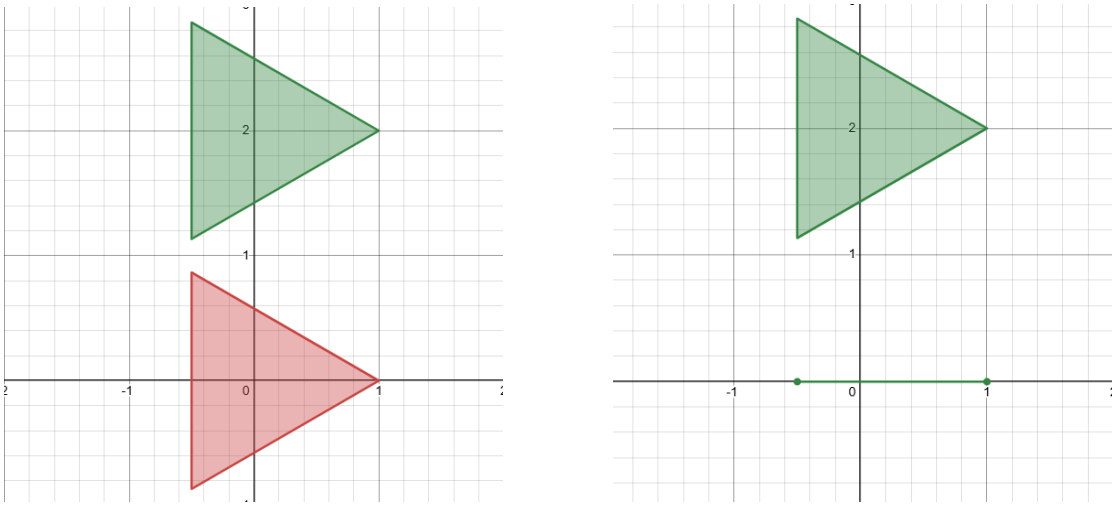


Figure 3.2: $W(A)$ vs. $W(B)$ vs. $W(\mathfrak{Re}(B))$

Example 3.5. Let $T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Since A and T are unitarily similar by the unitary matrix $U = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, their numerical ranges are equal as shown in the figure below.

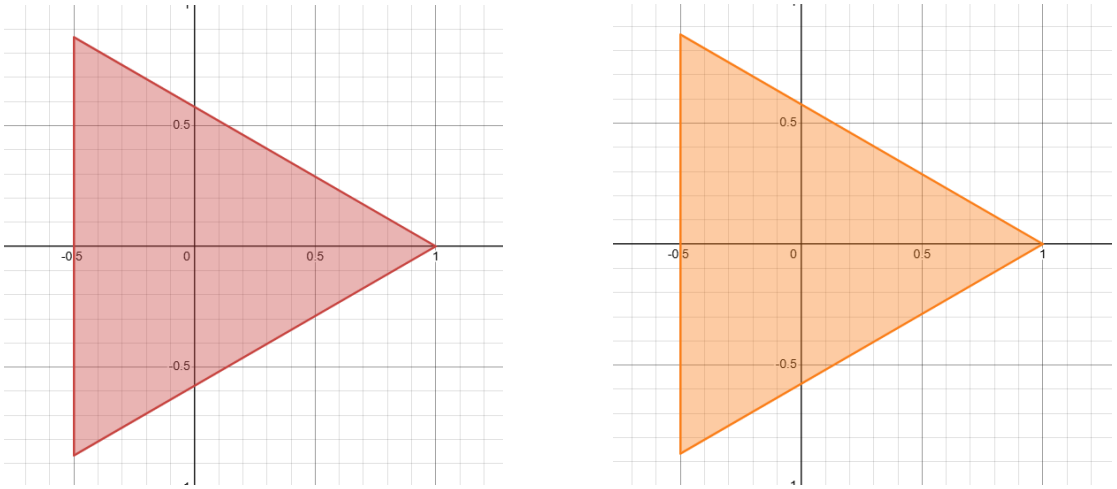


Figure 3.3: $W(A)$ vs. $W(T)$

In this chapter, we also establish the compactness and convexity of the numerical range.

Proposition 3.2. *[11, Property 1.2.1] $W(A)$ is compact for any $A \in M_n(\mathbb{C})$.*

Proof. Let $A \in M_n(\mathbb{C})$. Note that the set $W(A)$ is the image of the compact set $\{x \in \mathbb{C}^n \text{ is a unit vector}\}$ under the function $x \mapsto x^*Ax$, which is continuous by Corollary 2.16. By Proposition 2.17, it follows that $W(A)$ is compact. □

To establish the convexity of the numerical range, we use the following lemma.

Lemma 3.3 (Elliptical Range Theorem)**.** *[21] Let $A \in M_2(\mathbb{C})$. Let $\sigma(A) = \{\lambda_1, \lambda_2\}$. Then $W(A)$ is the elliptical disk centered at $\frac{1}{2}tr(A)$, has foci λ_1 and λ_2 and has minor axis of length equal to $\sqrt{tr(A^*A) - |\lambda_1|^2 - |\lambda_2|^2}$.*

A more general result is provided by the Toeplitz-Hausdorff Theorem.

Theorem 3.4. *[11, Section 1.3] $W(A)$ is convex for any $A \in M_n(\mathbb{C})$.*

Outline of the proof. The proof is divided into two parts:

1. *Reduction of the problem to the 2-by-2 case.* It suffices to show that the numerical range of a 2-by-2 matrix is convex (since the notion of convexity considers only pairs of vectors in $\mathbb{C}^n$).

2. *Demonstration of the convexity of the 2-by-2 case.* This will be accomplished by using the elliptical range theorem.

As a consequence of Theorem 3.4, we obtain some important properties of the numerical range in addition to Proposition 3.1.

Proposition 3.5 (Normality)**.** *[11, Property 1.2.9] Let $A \in M_n(\mathbb{C})$. If A is normal and $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$, then $W(A) = co(\sigma(A))$.*

Proof. Suppose A is normal and $\sigma(A) = \{\lambda_1, \ldots, \lambda_n\}$. By the spectral theorem and the unitary similarity invariance of $W(A)$, we may assume, without loss of generality, that A is a diagonal matrix.

For the forward inclusion, let $x = [x_1 \ \cdots \ x_n]^\top \in \mathbb{C}^n$ be a unit vector. Then $|x_j|^2 \geq 0$ for each j, and $\sum_{j=1}^{n} |x_j|^2 = 1$. Since $x^*Ax = \sum_{j=1}^{n} \overline{x_j} x_j \lambda_j = \sum_{j=1}^{n} |x_j|^2 \lambda_j$ is a convex combination of $\lambda_1, \ldots, \lambda_n$, it follows that $W(A) \subseteq \text{co}(\sigma(A))$.

For the reverse inclusion, recall that $\sigma(A) \subseteq W(A)$. Since $W(A)$ is convex, it follows from the definition of a convex hull that $\text{co}(\sigma(A)) \subseteq W(A)$.

Therefore, $W(A) = \text{co}(\sigma(A))$. □

As a consequence, we have the following corollary.

Corollary 3.6. *Let $A \in M_n(\mathbb{C})$. If A is Hermitian, then $W(A) = [\lambda_n(A),\ \lambda_1(A)]$.*

Proof. Since the eigenvalues of a Hermitian matrix are real, by definition,

$$\operatorname{co}(\sigma(A)) = [\lambda_n(A),\ \lambda_1(A)].$$

The conclusion follows from Proposition 3.5 since Hermitian matrices are normal. □

The notion of convexity plays a crucial role in characterizing the numerical range as the intersection of a family of closed half-planes.

Theorem 3.7. *Let $A \in M_n(\mathbb{C})$.*

$$W(A) = \bigcap_{\theta\in[0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}\left(e^{i\theta}v\right) \le \lambda_1\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right\}.$$

Proof. Let $w \in W(A)$. Fix $\theta \in [0, 2\pi)$. By Proposition 3.1

$$\mathfrak{Re}(e^{i\theta}w) \in \mathfrak{Re}(e^{i\theta}(W(A))) = \mathfrak{Re}(W(e^{i\theta}A)) = W\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right).$$

Since $\mathfrak{Re}\left(e^{i\theta}A\right)$ is Hermitian,

$$W\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) = \left[\lambda_n\left(\mathfrak{Re}(e^{i\theta}A)\right),\ \lambda_1\left(\mathfrak{Re}(e^{i\theta}A)\right)\right].$$

In particular, $\mathfrak{Re}(e^{i\theta}w) \le \lambda_1\left(\mathfrak{Re}(e^{i\theta}A)\right)$. Hence,

$$W(A) \subseteq \bigcap_{\theta\in[0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}\left(e^{i\theta}v\right) \le \lambda_1\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right\}.$$

Now, to establish the backward inclusion, we will prove its contrapositive, that is, for any $w \notin W(A)$,

$$w \notin \bigcap_{\theta\in[0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}\left(e^{i\theta}v\right) \le \lambda_1\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right\}.$$

Indeed, let $w \notin W(A)$. Since $W(A)$ is a closed convex set, Corollary 2.24 guarantees that there exists $\varphi \in [0, 2\pi)$ such that for all $z \in W(A)$,

$$\begin{aligned}
\mathfrak{Re}(z)\cos\varphi - \mathfrak{Im}(z)\sin\varphi &= [\cos\varphi \quad -\sin\varphi]\begin{bmatrix}\mathfrak{Re}(z)\\ \mathfrak{Im}(z)\end{bmatrix}\\
&< [\cos\varphi \quad -\sin\varphi]\begin{bmatrix}\mathfrak{Re}(w)\\ \mathfrak{Im}(w)\end{bmatrix}\\
&= \mathfrak{Re}(w)\cos\varphi - \mathfrak{Im}(w)\sin\varphi.
\end{aligned}$$

Note that for all $\mu \in \mathbb{C}$ and $\theta \in [0, 2\pi)$,

$$\begin{aligned}\mathfrak{Re}(e^{i\theta}\mu) &= \mathfrak{Re}((\cos\theta + i\sin\theta)\mu) \\ &= \mathfrak{Re}(\mu\cos\theta + i\mu\sin\theta) \\ &= \mathfrak{Re}(\mu\cos\theta) + \mathfrak{Re}(i\mu\sin\theta) \\ &= \mathfrak{Re}(\mu)\cos\theta - \mathfrak{Im}(\mu)\sin\theta.\end{aligned}$$

Thus, $\mathfrak{Re}(e^{i\varphi}z) < \mathfrak{Re}(e^{i\varphi}w)$ for all $z \in W(A)$. Now, let $x \in \mathbb{C}^n$ be a unit eigenvector corresponding to $\lambda_1(\mathfrak{Re}(e^{i\varphi}(A)))$. Then, $x^*Ax \in W(A)$ and so

$$\begin{aligned}\lambda_1(\mathfrak{Re}(e^{i\varphi}A)) &= x^*\mathfrak{Re}(e^{i\varphi}A)x \\ &= \mathfrak{Re}\left(x^*(e^{i\varphi}A)x\right) \\ &= \mathfrak{Re}\left(e^{i\varphi}x^*Ax\right) \\ &< \mathfrak{Re}(e^{i\varphi}w).\end{aligned}$$

Therefore, $w \notin \left\{v \in \mathbb{C} : \mathfrak{Re}(e^{i\varphi}v) \leq \lambda_1(\mathfrak{Re}(e^{i\varphi}A))\right\}$. In particular, we have

$$w \notin \bigcap_{\theta\in[0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}(e^{i\theta}v) \leq \lambda_1(\mathfrak{Re}(e^{i\theta}A))\right\}$$

and so the claim follows. □

Chapter 4

The k-Numerical Range

Several generalizations of the numerical range exist in the literature. In this study, we consider the k-numerical range of a matrix.

Definition 4.1. Let $k \in \{1, \ldots, n\}$. The *k-numerical range* of $A \in M_n(\mathbb{C})$ is the set

$$W_k(A) = \left\{ \sum_{j=1}^{k} x_j^* A x_j : \{x_1, \ldots, x_k\} \subset \mathbb{C}^n \text{ is orthonormal} \right\}.$$

Observe that $W_k(A) = W(A)$ when $k = 1$. In the following proposition, the k-numerical range is characterized using projections.

Proposition 4.1. *Let $k \in \{1, \ldots, n\}$. For any $A \in M_n(\mathbb{C})$,*

$$\begin{aligned} W_k(A) &= \{tr(V^*AV) : V \in M_{n,k}(\mathbb{C}) \text{ such that } V^*V = I_k\} \\ &= \{tr(AP) : P \in M_n(\mathbb{C}) \text{ is a projection of rank } k\} \\ &= \{tr(AU(I_k \oplus 0_{n-k})U^*) : U \in M_n(\mathbb{C}) \text{ is unitary}\}. \end{aligned}$$

Proof. By definition,

$$\begin{aligned} W_k(A) &= \left\{ \sum_{j=1}^{k} x_j^* A x_j : \{x_1, \ldots, x_k\} \subset \mathbb{C}^n \text{ is orthonormal} \right\} \\ &= \{\operatorname{tr}(V^*AV) : V \in M_{n,k}(\mathbb{C}) \text{ such that } V^*V = I_k\}. \end{aligned}$$

By Proposition 2.7,

$$\begin{aligned} W_k(A) &= \{\operatorname{tr}(AVV^*) : V \in M_{n,k}(\mathbb{C}) \text{ such that } V^*V = I_k\} \\ &= \{\operatorname{tr}(AP) : P \in M_n(\mathbb{C}) \text{ is a projection of rank } k\} \\ &= \{\operatorname{tr}(AU(I_k \oplus 0_{n-k})U^*) : U \in M_n(\mathbb{C}) \text{ is unitary}\}. \end{aligned}$$

□

The following properties of the k-numerical range can be seen in [15, Lemma 2.2] and [6, Theorem 1.1].

Proposition 4.2. *Let $k \in \{1, \ldots, n\}$. Let $A \in M_n(\mathbb{C})$. The following hold:*

1. $W_k(\alpha I_n + A) = k\alpha + W_k(A)$ *for any* $\alpha \in \mathbb{C}$; (*translation*)
2. $W_k(\beta A) = \beta W_k(A)$ *for any* $\beta \in \mathbb{C}$; (*scalar multiplication*)
3. $W_k(\mathfrak{Re}(A)) = \mathfrak{Re}(W_k(A))$; (*projection*)
4. $W_k(U^*AU) = W_k(A)$ *for any unitary* $U \in M_n(\mathbb{C})$; (*unitary invariance*)
5. $W_n(A) = \{tr(A)\}$.

Proof. Let $k \in \{1, \ldots, n\}$. Let $A \in M_n(\mathbb{C})$. For the first three parts, these are easily observed from the summation definition, or as in the proof of Proposition 3.1, the classical numerical range.

For the fourth part, let $U \in M_n(\mathbb{C})$ be unitary. If $P \in M_n(\mathbb{C})$ is a projection of rank k, then by the last part of Proposition 2.6, UPU^* is also a projection of rank k and so

$$\begin{aligned} W_k(U^*AU) &= \{\mathrm{tr}(U^*AUP) : P \in M_n(\mathbb{C}) \text{ is a projection of rank } k\} \\ &= \{\mathrm{tr}(AUPU^*) : P \in M_n(\mathbb{C}) \text{ is a projection of rank } k\} \\ &= \{\mathrm{tr}(AQ) : Q \in M_n(\mathbb{C}) \text{ is a projection of rank } k\} \\ &= W_k(A). \end{aligned}$$

For the last part, note that

$$\begin{aligned} W_n(A) &= \{\mathrm{tr}(AP) : P \in M_n(\mathbb{C}) \text{ is a projection of rank } n\} \\ &= \{\mathrm{tr}(AI_n)\} \\ &= \{\mathrm{tr}(A)\} \end{aligned}$$

by the second part of Proposition 2.8. □

Proposition 4.3. *[6, Theorem 1.1(ii)] Let $A \in M_n(\mathbb{C})$. Then $W_k(A) = tr(A) - W_{n-k}(A)$ for any $k \in \{1, \ldots, n-1\}$.*

Proof. Let $k \in \{1, \ldots, n-1\}$. By Proposition 4.1,

$$\begin{aligned}
W_k(A) &= \{\operatorname{tr}(AU(I_k \oplus 0_{n-k})U^*) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \{\operatorname{tr}(A(U(I_k \oplus 0_{n-k} + 0_k \oplus I_{n-k} - 0_k \oplus I_{n-k})U^*)) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \{\operatorname{tr}(A(U(I_n)U^* - U(0_k \oplus I_{n-k})U^*)) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \{\operatorname{tr}(A(I_n - U(0_k \oplus I_{n-k})U^*)) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \{\operatorname{tr}(A - AU(0_k \oplus I_{n-k})U^*) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \{\operatorname{tr}(A) - \operatorname{tr}(AU(0_k \oplus I_{n-k})U^*) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \operatorname{tr}(A) - \{\operatorname{tr}(AU(0_k \oplus I_{n-k})U^*) : U \in M_n(\mathbb{C}) \text{ is unitary}\} \\
&= \operatorname{tr}(A) - W_{n-k}(A).
\end{aligned}$$

□

Proposition 4.3 implies that it is enough to study the k-numerical range for $k \in \{1, \ldots, \lfloor \frac{n}{2} \rfloor\}$ since the k-numerical range for other values of k are already determined by the above relation. Similar to Proposition 3.2, we establish the compactness of the k-numerical range.

Proposition 4.4. *Let $k \in \{1, \ldots, n\}$. Then $W_k(A)$ is compact for any $A \in M_n(\mathbb{C})$.*

Proof. Let $A \in M_n(\mathbb{C})$ and let $\mathscr{U}$ and $\mathscr{P}_k$ be the set of all unitary matrices in $M_n(\mathbb{C})$ and the set of all projections of rank k in $M_n(\mathbb{C})$, respectively. Define $D = I_k \oplus 0_{n-k}$ and consider the functions

$$\begin{aligned}
f : \mathscr{U} &\longrightarrow M_n(\mathbb{C}) \\
U &\longmapsto UDU^*
\end{aligned}$$

and

$$\begin{aligned}
g : \mathscr{P}_k &\longrightarrow \mathbb{C} \\
P &\longmapsto \operatorname{tr}(AP).
\end{aligned}$$

By Propositions 2.7 and 4.1, $f(\mathscr{U}) = \mathscr{P}_k$ and $g(\mathscr{P}_k) = W_k(A)$, respectively. Thus, $g(f(\mathscr{U})) = g(\mathscr{P}_k) = W_k(A)$. Since $\mathscr{U}$ is compact [12, Section 2.1], it suffices to show that $h = g \circ f$ is continuous on $\mathscr{U}$. Indeed, suppose $\{U_j\}_{j=1}^{\infty}$ is a sequence of unitary matrices in $M_n(\mathbb{C})$ where $\lim_{j\to\infty} U_j = U$ for some $U \in \mathscr{U}$. By Corollary 2.14, it follows that

$$\lim_{j\to\infty} f(U_j) = \lim_{j\to\infty} U_j D U_j^* = \left(\lim_{j\to\infty} U_j\right) D \left(\lim_{j\to\infty} U_j^*\right) = UDU^* = f(U),$$

and so f is continuous on $\mathscr{U}$. Likewise, g is continuous on $\mathscr{P}_k$ since if $\{P_j\}_{j=1}^{\infty}$ is a sequence of projections of rank k in $M_n(\mathbb{C})$ such that $\lim_{j\to\infty} P_j = P$ for some $P \in M_n(\mathbb{C})$, then

$$\begin{aligned}
\lim_{j\to\infty} g(P_j) &= \lim_{j\to\infty} \operatorname{tr}(AP_j) \\
&= \lim_{j\to\infty} \sum_{i=1}^{n} e_i^* A P_j e_i \\
&= \sum_{i=1}^{n} \lim_{j\to\infty} e_i^* A P_j e_i \\
&= \sum_{i=1}^{n} e_i^* A \left(\lim_{j\to\infty} P_j \right) e_i \\
&= \sum_{i=1}^{n} e_i^* A P e_i \\
&= \operatorname{tr}(AP) \\
&= g(P).
\end{aligned}$$

Therefore, $h = g \circ f$ is continuous on $\mathscr{U}$. By Proposition 2.17, $W_k(A) = h(\mathscr{U})$ is compact. □

In the remainder of this chapter, we discuss results that lead up to the proof of the convexity of the k-numerical range.

Lemma 4.5. *Let $P, Q \in M_n(\mathbb{C})$ be projections of rank k. Then there exist orthonormal bases $\{x_i\}_{i=1}^k$ and $\{y_i\}_{i=1}^k$ for $Im(P)$ and $Im(Q)$, respectively, such that $y_i^* x_j = 0$ for all distinct $i, j \in \{1, \ldots, k\}$.*

Proof. Suppose P and Q are projections of rank k. By Proposition 2.7, we have $P = VV^*$ and $Q = WW^*$ for some $V, W \in M_{n,k}(\mathbb{C})$, where $V^*V = W^*W = I_k$. By Theorem 2.10, there exist unitary $G, F \in M_k(\mathbb{C})$ and $\Sigma = \operatorname{diag}(\alpha_1, \ldots, \alpha_k) \in M_k(\mathbb{C})$ such that $W^*V = G\Sigma F^*$. For all $i \in \{1, \ldots, k\}$, let $x_i = VFe_i$ and $y_i = WGe_i$. Then

$$Px_i = PVFe_i = VFe_i = x_i \text{ and } Qy_i = QWGe_i = WGe_i = y_i,$$

that is, $x_i \in \operatorname{Im}(P)$ and $y_i \in \operatorname{Im}(Q)$. In addition,

$$x_i^* x_j = (VFe_i)^* VFe_j = e_i^* F^* V^* VFe_j = e_i^* F^* Fe_j = e_i^* e_j = \delta_{ij}$$

and

$$y_i^* y_j = (WGe_i)^* WGe_j = e_i^* G^* W^* WGe_j = e_i^* G^* Ge_j = e_i^* e_j = \delta_{ij}$$

where $\delta_{ij} = 0$ if $i \neq j$ and $\delta_{ij} = 1$ if $i = j$. By Proposition 2.4, this implies that $\{x_j\}_{j=1}^k$ and $\{y_j\}_{j=1}^k$ are linearly independent sets. Since Σ is diagonal,

$$y_i^* x_j = (WGe_i)^* VFe_j = e_i^* G^* W^* VFe_j = e_i^* G^* G\Sigma F^* Fe_j = e_i^* \Sigma e_j = 0$$

for distinct $i, j \in \{1, \ldots, k\}$. Since P has rank k and $\{x_j\}_{j=1}^k$ is a linearly independent set, it follows that $\{x_j\}_{j=1}^k$ is basis for $\mathrm{Im}(P)$. The fact that $\{y_j\}_{j=1}^k$ is a basis for $\mathrm{Im}(Q)$ can be shown similarly. □

The proof of the previous result in [1, Chapter 12] is different from the one presented here. The above proof uses ideas from matrix decompositions.

Lemma 4.6. *[11, Problem 1.3.1] Let $x, y \in \mathbb{C}^n$ be unit vectors. There exists unitary $U \in M_n(\mathbb{C})$ and $v, w \in \mathbb{C}^n$ such that $x = Uv$ and $y = Uw$, and the last $n-2$ entries of v and w are equal to* 0.

Proof. Suppose x and y are linearly dependent. Then $y = \alpha x$ for some $\alpha \in \mathbb{C}$. By extending $\{x\}$ to an orthonormal basis $\{x, x_2, \ldots, x_n\}$ for $\mathbb{C}^n$ and by letting $U = [x \;\; x_2 \;\; \cdots \;\; x_n]$ (which is unitary), we have

$$U^* x = [\overline{x} \;\; \overline{x_2} \;\; \cdots \;\; \overline{x_n}]^\top x = [x^\top \overline{x} \;\; x^\top \overline{x_2} \;\; \cdots \;\; x^\top \overline{x_n}]^\top = [1 \;\; 0 \;\; \cdots \;\; 0]^\top = e_1.$$

The claim follows if we set $v = U^* x$ and $w = \alpha v = \alpha U^* x = U^*(\alpha x) = U^* y$.

On the other hand, suppose x and y are linearly independent. By the Gram-Schmidt process, there exists an orthonormal basis $\{u_1, u_2\}$ for $\mathrm{span}\{x, y\}$. In particular, $x = au_1 + bu_2$ and $y = cu_1 + du_2$ for some $a, b, c, d \in \mathbb{C}$. Extend $\{u_1, u_2\}$ to an orthonormal basis $\{u_1, u_2, u_3, \ldots, u_n\}$ for $\mathbb{C}^n$ and form the unitary matrix $U = [u_1 \;\; u_2 \;\; u_3 \;\; \cdots \;\; u_n] \in M_n(\mathbb{C})$ so that

$$\begin{aligned} U^* x &= [u_1 \;\; u_2 \;\; u_3 \;\; \cdots \;\; u_n]^* (au_1 + bu_2) \\ &= \begin{bmatrix} au_1^* u_1 + bu_1^* u_2 \\ au_2^* u_1 + bu_2^* u_2 \\ au_3^* u_1 + bu_3^* u_2 \\ \vdots \\ au_n^* u_1 + bu_n^* u_2 \end{bmatrix} \\ &= [a \;\; b \;\; 0 \;\; \cdots \;\; 0]^\top. \end{aligned}$$

In the same manner, $U^* y = ce_1 + de_2$. Taking $v = U^* x$ and $w = U^* y$ gives us the desired conclusion. □

Lemma 4.7. *Let $A \in M_n(\mathbb{C})$ and let $x, y \in \mathbb{C}^n$ be unit vectors. For all $t \in [0,1]$, there exists unit vector $z \in span\{x, y\}$ such that $tx^*Ax + (1-t)\, y^*Ay = z^*Az$.*

Proof. Fix $t \in [0,1]$. By Lemma 4.6, there exists unitary matrix $U \in M_n(\mathbb{C})$ where $x = Uv$ and $y = Uw$ with $v = [v_1 \;\; v_2 \;\; 0 \;\; \cdots \;\; 0]^\top$ and $w = [w_1 \;\; w_2 \;\; 0 \;\; \cdots \;\; 0]^\top$. Since U is unitary and both x and y are unit vectors, v and w are unit vectors, that is, $|v_1|^2 + |v_2|^2 = |w_1|^2 + |w_2|^2 = 1$. Thus, $\xi = [v_1 \;\; v_2]^\top, \eta = [w_1 \;\; w_2]^\top \in \mathbb{C}^2$ are unit vectors and so if $B = U^*AU \in M_n(\mathbb{C})$ and $B_1 = [e_1 \;\; e_2]^\top B[e_1 \;\; e_2] \in M_2(\mathbb{C})$, then

$$\begin{aligned} tx^*Ax + (1-t)\, y^*Ay &= tv^*U^*AUv + (1-t)\, w^*U^*AUw \\ &= tv^*Bv + (1-t)\, w^*Bw \\ &= t\xi^*B_1\xi + (1-t)\, \eta^*B_1\eta. \end{aligned}$$

If ξ and η are linearly dependent, then there exists $\alpha \in \mathbb{C}$ such that $\eta = \alpha\xi$. Since η is a unit vector, we have $|\alpha| = 1$. Thus,

$$\begin{aligned} t\xi^*B_1\xi + (1-t)\, \eta^*B_1\eta &= t\xi^*B_1\xi + (1-t)\, |\alpha|^2\, \xi^*B_1\xi \\ &= t\xi^*B_1\xi + \xi^*B_1\xi - t\xi^*B_1\xi \\ &= \xi^*B_1\xi. \end{aligned}$$

By taking $z = x \in \operatorname{span}\{x, y\}$, we have

$$\begin{aligned} tx^*Ax + (1-t)\, y^*Ay &= \xi^*B_1\xi \\ &= V^*BV \\ &= x^*UBU^*x \\ &= z^*Az. \end{aligned}$$

Now, if ξ and η are linearly independent, then $\operatorname{span}\{\xi, \eta\} = \mathbb{C}^2$. Since $W(B_1)$ is convex, it follows from the definition that there exists a unit vector $\zeta = [z_1 \;\; z_2]^\top \in \mathbb{C}^2$ such that $t\xi^*B_1\xi + (1-t)\, \eta^*B_1\eta = \zeta^*B_1\zeta$. Consider $z = U[z_1 \;\; z_2 \;\; 0 \;\; \cdots \;\; 0]^\top$ and observe that

$$\begin{aligned} tx^*Ax + (1-t)\, y^*Ay &= t\xi^*B_1\xi + (1-t)\, \eta^*B_1\eta \\ &= \zeta^*B_1\zeta \\ &= [\overline{z_1} \;\; \overline{z_2} \;\; 0 \;\; \cdots \;\; 0]B[z_1 \;\; z_2 \;\; 0 \;\; \cdots \;\; 0]^\top \\ &= z^*UBU^*z \\ &= z^*Az. \end{aligned}$$

Since $\zeta \in \mathbb{C}^2 = \operatorname{span}\{\xi, \eta\}$, we have $[z_1 \quad z_2 \quad 0 \quad \cdots \quad 0]^\top \in \operatorname{span}\{v, w\}$. In particular,

$$z = U[z_1 \quad z_2 \quad 0 \quad \cdots \quad 0]^\top \in \operatorname{span}\{Uv, Uw\} = \operatorname{span}\{x, y\}.$$

Since ζ is a unit vector and U is unitary,

$$\begin{aligned}
\|z\|_2^2 &= \|U[z_1 \quad z_2 \quad 0 \quad \cdots \quad 0]^\top\|_2^2 \\
&= [\overline{z_1} \quad \overline{z_2} \quad 0 \quad \cdots \quad 0]U^*U[z_1 \quad z_2 \quad 0 \quad \cdots \quad 0]^\top \\
&= [\overline{z_1} \quad \overline{z_2} \quad 0 \quad \cdots \quad 0][z_1 \quad z_2 \quad 0 \quad \cdots \quad 0]^\top \\
&= \|[z_1 \quad z_2 \quad 0 \quad \cdots \quad 0]^\top\|_2^2 \\
&= \|\zeta\|_2^2 \\
&= 1,
\end{aligned}$$

that is, z is a unit vector. Therefore, there exists unit vector $z \in \operatorname{span}\{x, y\}$ such that $tx^*Ax + (1-t)\, y^*Ay = z^*Az$. □

Theorem 4.8. *[9, Solution 211], [1] Let $k \in \{1, \ldots, n\}$. Then $W_k(A)$ is convex for any $A \in M_n(\mathbb{C})$.*

Proof. Let $A \in M_n(\mathbb{C})$ and suppose $P, Q \in M_n(\mathbb{C})$ are projections of rank k. Then $\operatorname{tr}(AP) \in W_k(A)$. Now, fix $t \in [0,1]$. By Lemma 4.5, there exist orthonormal $\{x_i\}_{i=1}^k$ and $\{y_i\}_{i=1}^k$ spanning $\operatorname{Im}(P)$ and $\operatorname{Im}(Q)$, respectively, where $y_j^* x_i = 0$ for all distinct $i, j \in \{1, \ldots, k\}$. Moreover, $x_i^* A x_i, y_i^* A y_i \in W(A)$ since x_i and y_i are unit vectors for all $i \in \{1, \ldots, k\}$.

Now, let $t \in [0,1]$ and $i \in \{1, \ldots, k\}$. By Lemma 4.7, there exists a unit vector $z_i \in \operatorname{span}\{x_i, y_i\}$ such that $tx_i^*Ax_i + (1-t)\, y_i^*Ay_i = z_i^*Az_i$. In particular, $z_i = \alpha_i x_i + \beta_i y_i$ for some $\alpha_i, \beta_i \in \mathbb{C}$ and

$$t\sum_{i=1}^k x_i^*Ax_i + (1-t)\sum_{i=1}^k y_i^*Ay_i = \sum_{i=1}^k z_i^*Az_i.$$

We want to show that the above summations are in $W_k(A)$. For the right-hand side, $\{x_i\}_{i=1}^k$ and $\{y_i\}_{i=1}^k$ are orthonormal with $y_j^*x_i = 0 = (y_j^*x_i)^* = x_i^*y_j$ for all distinct $i, j \in \{1, \ldots, k\}$, so

$$\begin{aligned}
z_j^* z_i &= \left(\overline{\alpha_j} x_j^* + \overline{\beta_j} y_j^*\right)(\alpha_i x_i + \beta_i y_i) \\
&= \overline{\alpha_j}\alpha_i x_j^* x_i + \overline{\alpha_j}\beta_i x_j^* y_i + \overline{\beta_j}\alpha_i y_j^* x_i + \overline{\beta_j}\beta_i y_j^* y_i \\
&= 0.
\end{aligned}$$

Hence, $\{z_i\}_{i=1}^k$ is orthonormal. Moreover, if $Z = [z_1 \quad \cdots \quad z_k] \in M_{n,k}(\mathbb{C})$, then $Z^*Z = I_k$ and $R = ZZ^* \in M_n(\mathbb{C})$ is a projection of rank k. Thus,

$$\operatorname{tr}(AR) = \operatorname{tr}(AZZ^*) = \operatorname{tr}(Z^*AZ) = \sum_{i=1}^{k} z_i^* A z_i \in W_k(A).$$

For the left-hand summations, since P is a projection of rank k, there exists $V \in M_{n,k}(\mathbb{C})$ such that $V^*V = I_k$ and $P = VV^*$. Thus, $\operatorname{tr}(AP) = \operatorname{tr}(AVV^*) = \operatorname{tr}(V^*AV)$. Now, define $X = [x_1 \quad \cdots \quad x_k] \in M_{n,k}(\mathbb{C})$. Since $x_1, \ldots, x_k$ are orthonormal, $X^*X = I_k$. By Proposition 2.6, $PX = X$. Thus,

$$(V^*X)^* V^*X = X^*VV^*X = X^*PX = X^*X = I_k.$$

Since $V^*X \in M_k(\mathbb{C})$, this implies that V^*X is unitary and so

$$\begin{aligned}\sum_{i=1}^{k} x_i^* A x_i &= \operatorname{tr}(X^*AX) \\ &= \operatorname{tr}(X^*PAPX) \\ &= \operatorname{tr}(X^*VV^*AVV^*X) \\ &= \operatorname{tr}(V^*AV) \\ &= \operatorname{tr}(AP).\end{aligned}$$

By a similar reasoning, $\operatorname{tr}(AQ) = \sum_{i=1}^{k} y_i^* A y_i$. Hence,

$$t\cdot\operatorname{tr}(AP)+(1-t)\cdot\operatorname{tr}(AQ) = t\sum_{i=1}^{k} x_i^*Ax_i+(1-t)\sum_{i=1}^{k} y_i^*Ay_i = \sum_{i=1}^{k} z_i^*Az_i = \operatorname{tr}(AR).$$

Therefore, $W_k(A)$ is convex. □

Similar to Proposition 3.5, the normality of the k-numerical range can be deduced from its convexity. We first consider the following lemma.

Lemma 4.9. *[6, Lemma 1.3] Let $n \in \mathbb{N}$ and $k \in \{1, \ldots, n\}$. The set*

$$\rho_k = \left\{ [p_1 \quad \cdots \quad p_n]^\top \in \mathbb{R}^n : 0 \le p_i \le 1 \text{ for all } i \in \{1, \ldots, n\} \text{ and } \sum_{i=1}^{n} p_i = k \right\}$$

is convex and compact. Moreover, an extreme point of ρ_k has exactly k entries equal to 1 and the rest are equal to 0.

Proof. Fix $k \in \{1, \ldots, n\}$. Note that ρ_k is nonempty since it contains vectors in $\mathbb{R}^n$ with exactly k entries equal to 1 and the remaining $n-k$ entries equal to 0. Now, let $p = [p_1 \ \cdots \ p_n]^\top \in \rho_k$ and $q = [q_1 \ \cdots \ q_n]^\top \in \rho_k$. Then for all $t \in [0,1]$ and all $i \in \{1, \ldots, n\}$,

$$0 \leq \min\{p_i,\ q_i\} \leq tp_i + (1-t)\, q_i \leq \max\{p_i, q_i\} \leq 1$$

and

$$t\sum_{i=1}^{n} p_i + (1-t)\sum_{i=1}^{n} q_i = tk + (1-t)\, k = k.$$

Thus, $tp + (1-t)\, q \in \rho_k$ for all $t \in [0,1]$ and so ρ_k is convex.

We now show that ρ_k is compact. Since $p \in \rho_k$, observe that

$$k^2 = \left(\sum_{i=1}^{n} p_i\right)^2 = \sum_{i=1}^{n} p_i^2 + 2\sum_{i \neq j}^{n} p_i p_j.$$

Since $p_i \geq 0$ for all $i \in \{1, \ldots, n\}$,

$$\sum_{i=1}^{n} p_i^2 = k^2 - 2\sum_{i \neq j}^{n} p_i p_j \leq k^2.$$

Thus, $\|p\|_2 \leq k$ with $k > 0$ and so ρ_k is bounded. Moreover, let $\{r^{(\ell)}\}_{\ell=1}^{\infty}$ be a convergent sequence of elements of ρ_k where $r^{(\ell)} = [r_{1,\ell} \ \cdots \ r_{n,\ell}]^\top$ for all $\ell \in \mathbb{N}$. Then $\lim_{\ell \to \infty} r^{(\ell)} = r$ for some $r = [r_1 \ \cdots \ r_n]^\top \in \mathbb{R}^n$. In particular, by Proposition 2.13, $\lim_{\ell \to \infty} r_{j,\ell} = r_j$ for each $j \in \{1, \ldots, n\}$. Moreover, since $0 \leq r_{j,\ell} \leq 1$ for all $j \in \{1, \ldots, n\}$ and $\ell \in \mathbb{N}$, we have $0 \leq r_j \leq 1$ for all $j \in \{1, \ldots, n\}$. Furthermore, since $\sum_{j=1}^{n} r_{j,\ell} = k$ for all $\ell \in \mathbb{N}$,

$$\sum_{j=1}^{n} r_j = \sum_{j=1}^{n} \lim_{\ell \to \infty} r_{j,\ell} = \lim_{\ell \to \infty} \sum_{j=1}^{n} r_{j,\ell} = \lim_{\ell \to \infty} k = k.$$

Thus, $r \in \rho_k$ and so ρ_k is closed. Hence, ρ_k is compact and so ρ_k has an extreme point by Proposition 2.21.

Fix $s \in \{1, \ldots, n\}$. Suppose $0 < p_s < 1$. Then $p_s \notin \mathbb{Z}$. Since $\sum_{i=1}^{n} p_i = k \in \mathbb{Z}$, it follows that $0 < p_j < 1$ for some $j \neq s$. Without loss of generality, assume that $s < j$. Consider $\varepsilon = \min\{p_s, p_j, 1-p_s, 1-p_j\} > 0$ and let

$$p' = [p_1 \ \cdots \ p_s - \varepsilon \ \cdots \ p_j + \varepsilon \ \cdots \ p_n]^\top$$

and

$$p'' = [p_1 \ \cdots \ p_s + \varepsilon \ \cdots \ p_j - \varepsilon \ \cdots \ p_n]^\top.$$

Notice that $p = \frac{1}{2}(p' + p'')$, and that

$$0 = p_s - p_s \leq p_s - \varepsilon < p_s < 1 \text{ and } 0 < p_s < p_s + \varepsilon \leq p_s + 1 - p_s = 1.$$

Similarly,

$$0 \leq p_j - \varepsilon < 1 \text{ and } 0 < p_j + \varepsilon \leq 1.$$

Moreover,

$$p_s - \varepsilon + p_j + \varepsilon + \sum_{i \neq s,j} p_i = \sum_{i=1}^{n} p_i = k.$$

Hence, $p', p'' \in \rho_k$. Now, since $\frac{1}{2} \in (0,1)$ and either p' or p'' is not equal to p, it follows that p is not an extreme point of ρ_k. Thus, if p is an extreme point of ρ_k, then $p_i \in \{0,1\}$ for all $i \in \{1,\ldots,n\}$. Since $\sum_{i=1}^{n} p_i = k$, then p has exactly k entries equal to 1 while the rest are equal to 0. □

Let $n \in \mathbb{N}$ and let $k \in \{1,\ldots,n\}$. By Krein-Milman theorem, $\rho_k = \text{co}(\text{Ext}(\rho_k))$. In addition, the cardinality of ρ_k is at most $\binom{n}{k}$ since an extreme point of ρ_k has exactly n entries, k of which are equal to 1. Thus, $\text{Ext}(\rho_k)$ is finite.

Proposition 4.10 (Normality)**.** *[6, Corollary 2] Let $k \in \{1,\ldots,n\}$. Let $A \in M_n(\mathbb{C})$. If A is normal and $\sigma(A) = \{\lambda_1,\ldots,\lambda_n\}$, then*

$$W_k(A) = co\left\{\sum_{i=1}^{k} \lambda_{\tau(i)} : \tau \in S_n\right\}.$$

Proof. Fix $k \in \{1,\ldots,n\}$. We first show the forward inclusion. Let $v \in W_k(A)$. Then $v = \text{tr}(AP)$ for some projection P of rank k. Since A is normal, Theorem 2.9 implies $A = \sum_{j=1}^{n} \lambda_j E_j$, where E_j is a projection of rank 1 for all $j \in \{1,\ldots,n\}$, and $\sum_{j=1}^{n} E_j = I_n$.

Since E_j is a projection of rank 1, by Proposition 2.7, there exists a unit vector $x_j \in \mathbb{C}^n$ such that $E_j = x_j x_j^*$ for each $j \in \{1,\ldots,n\}$. Now, observe that $x_j^* P x_j$ is Hermitian. Thus, by Corollary 3.6, $\text{tr}(E_j P) = \text{tr}(x_j x_j^* P) = x_j^* P x_j \in W(P) = [0,1]$. In addition,

$$\sum_{j=1}^{n} \text{tr}(E_j P) = \text{tr}\left(\sum_{j=1}^{n} E_j P\right) = \text{tr}\left(\left(\sum_{j=1}^{n} E_j\right) P\right) = \text{tr}(I_n P) = \text{tr}(P) = k.$$

Hence, $[\text{tr}(E_1 P) \quad \cdots \quad \text{tr}(E_n P)]^\top \in \rho_k$. Now, let m be the cardinality of $\text{Ext}(\rho_k)$. Since $\rho_k = \text{co}(\text{Ext}(\rho_k))$, there exist $a_i \in [0,1]$ and $p_i = [p_{i1} \quad \cdots \quad p_{in}]^\top \in \rho_k$ for

all $i \in \{1, \ldots, m\}$ such that $\sum_{i=1}^{m} a_i = 1$ and $[\mathrm{tr}(E_1 P) \quad \cdots \quad \mathrm{tr}(E_n P)]^\top = \sum_{i=1}^{m} a_i p_i$. In particular, $\mathrm{tr}(E_j P) = \sum_{i=1}^{m} a_i p_{ij}$ where for all $i \in \{1, \ldots, m\}$, there are exactly k p_{ij}'s that are equal to 1 while the rest are equal to 0. Thus,

$$\begin{aligned} v &= \mathrm{tr}(AP) \\ &= \mathrm{tr}\left(\sum_{j=1}^{n} \lambda_j E_j P\right) \\ &= \sum_{j=1}^{n} \lambda_j \mathrm{tr}(E_j P) \\ &= \sum_{j=1}^{n} \lambda_j \left(\sum_{i=1}^{m} a_i p_{ij}\right) \\ &= \sum_{j=1}^{n} \left(\sum_{i=1}^{m} \lambda_j a_i p_{ij}\right) \\ &= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} \lambda_j a_i p_{ij}\right) \\ &= \sum_{i=1}^{m} a_i \sum_{j=1}^{n} \lambda_j p_{ij} \end{aligned}$$

which shows that v is a convex combination of $\sum_{i=1}^{k} \lambda_{\tau(i)}$ for all $\tau \in S_n$. Thus, $W_k(A) \subseteq \mathrm{co}\left\{\sum_{i=1}^{k} \lambda_{\tau(i)} : \tau \in S_n\right\}$.

On the other hand, since A is normal, there exists orthonormal $\{x_i\}_{i=1}^{n} \subset \mathbb{C}^n$ such that $Ax_i = \lambda_i x_i$. In particular, $x_i^* A x_i = \lambda_i$ for all $i \in \{1, \ldots, n\}$ and so $\sum_{i=1}^{k} \lambda_{\tau(i)} = \sum_{i=1}^{k} x_{\tau(i)}^* A x_{\tau(i)} \in W_k(A)$ for all $\tau \in S_n$. Since $W_k(A)$ is convex, the reverse set inclusion follows. Therefore, $W_k(A) = \mathrm{co}\left\{\sum_{i=1}^{k} \lambda_{\tau(i)} : \tau \in S_n\right\}$. □

Similar to Corollary 3.6, we have the following result.

Corollary 4.11. *Let $k \in \{1, \ldots, n\}$ and $A \in M_n(\mathbb{C})$ be Hermitian. Then*

$$W_k(A) = \left[\sum_{j=1}^{k} \lambda_{n-k+j}(A), \ \sum_{j=1}^{k} \lambda_j(A)\right].$$

Proof. Suppose A is Hermitian. By Corollary 2.28,

$$\sum_{j=1}^{k} \lambda_{n-k+j}(A) \leq \sum_{j=1}^{k} \lambda_{\tau(j)}(A) \leq \sum_{j=1}^{k} \lambda_j(A).$$

Moreover, A is normal and so by Proposition 4.10,

$$W_k(A) = \operatorname{co}\left\{\sum_{j=1}^{k} \lambda_{\tau(j)}(A) : \tau \in S_n\right\} = \left[\sum_{j=1}^{k} \lambda_{n-k+j}(A),\ \sum_{j=1}^{k} \lambda_j(A)\right].$$

□

By establishing the above properties of the k-numerical range, we now have the following main result.

Theorem 4.12. *Let $k \in \{1, \ldots, n\}$. Let $A \in M_n(\mathbb{C})$. Then*

$$W_k(A) = \bigcap_{\theta \in [0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}\left(e^{i\theta}v\right) \leq \sum_{j=1}^{k} \lambda_j\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right\}.$$

Proof. Let $w \in W_k(A)$. Fix $\theta \in [0, 2\pi)$. By Proposition 4.2,

$$\mathfrak{Re}(e^{i\theta}w) \in \mathfrak{Re}(e^{i\theta}(W_k(A))) = \mathfrak{Re}(W_k(e^{i\theta}A)) = W_k\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right).$$

Since $\mathfrak{Re}\left(e^{i\theta}A\right)$ is Hermitian,

$$W_k\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) = \left[\sum_{j=1}^{k} \lambda_{n-k+j}\left(\mathfrak{Re}(e^{i\theta}A)\right),\ \sum_{j=1}^{k} \lambda_j\left(\mathfrak{Re}(e^{i\theta}A)\right)\right].$$

In particular, $\mathfrak{Re}(e^{i\theta}w) \leq \sum_{j=1}^{k} \lambda_j\left(\mathfrak{Re}(e^{i\theta}A)\right)$. Hence,

$$W_k(A) \subseteq \bigcap_{\theta \in [0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}\left(e^{i\theta}v\right) \leq \sum_{j=1}^{k} \lambda_j\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right\}.$$

We now establish the backward inclusion by contraposition, that is, for any $w \notin W_k(A)$,

$$w \notin \bigcap_{\theta \in [0,2\pi)} \left\{v \in \mathbb{C} : \mathfrak{Re}(e^{i\theta}v) \leq \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\theta}A))\right\}.$$

Indeed, let $w \notin W_k(A)$. Since $W_k(A)$ is a closed convex set, Corollary 2.24 guarantees that there exists $\varphi \in [0, 2\pi)$ such that $\mathfrak{Re}(e^{i\varphi}z) < \mathfrak{Re}(e^{i\varphi}w)$ for all $z \in W_k(A)$. Also, note that $\mathfrak{Re}(e^{i\varphi}A)$ is Hermitian. By Theorem 2.9, there exists an orthonormal $\{x_j\}_{j=1}^{k}$ such that x_j is an eigenvector of $\mathfrak{Re}(e^{i\varphi}A)$ corresponding to

$\lambda_j(\mathfrak{Re}(e^{i\varphi}A))$ for each $j \in \{1,\dots,k\}$. Using these, we get $\sum_{j=1}^{k} x_j^* A x_j \in W_k(A)$, and

$$\begin{aligned}
\sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}A)) &= \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}A)) x_j^* x_j \\
&= \sum_{j=1}^{k} x_j^* (\lambda_j(\mathfrak{Re}(e^{i\varphi}A)) x_j) \\
&= \sum_{j=1}^{k} x_j^* \mathfrak{Re}(e^{i\varphi}A) x_j \\
&= \mathfrak{Re}\left(\sum_{j=1}^{k} x_j^* (e^{i\varphi}A) x_j \right) \\
&= \mathfrak{Re}\left(e^{i\varphi} \sum_{j=1}^{k} x_j^* A x_j \right) \\
&< \mathfrak{Re}(e^{i\varphi} w).
\end{aligned}$$

Thus, $w \notin \left\{ v \in \mathbb{C} : \mathfrak{Re}(e^{i\varphi}v) \leq \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}A)) \right\}$. In particular, we have w is not in $\bigcap_{\theta\in[0,2\pi)} \left\{ v \in \mathbb{C} : \mathfrak{Re}(e^{i\theta}v) \leq \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\theta}A)) \right\}$ and so the claim follows. □

This result can be extended to c-numerical ranges, as shown in [17, Theorem 2.1]. The following is an immediate consequence of Theorem 4.12.

Corollary 4.13. *Let $k \in \{1,\dots,n\}$. Let $A, B \in M_n(\mathbb{C})$. Then $W_k(B) \subseteq W_k(A)$ if and only if for all $\theta \in [0, 2\pi)$,*

$$\sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\theta}B)) \leq \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\theta}A)).$$

Proof. Note that the converse is an immediate consequence of Theorem 4.12. Here, we show the forward implication by contraposition. Indeed, suppose that

$$\sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}B)) > \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}A))$$

for some $\varphi \in [0, 2\pi)$. Since $\mathfrak{Re}(e^{i\varphi}B)$ is Hermitian, there exists an orthonormal $\{y_j\}_{j=1}^{k}$ such that y_j is an eigenvector of $\mathfrak{Re}(e^{i\varphi}B)$ corresponding to $\lambda_j(\mathfrak{Re}(e^{i\varphi}B))$ for all $j \in \{1,\dots,k\}$. If $w = \sum_{j=1}^{k} y_j^* B y_j$, then by definition, $w \in W_k(B)$. However,

$w \notin W_k(A)$ since

$$\begin{aligned}
\mathfrak{Re}(e^{i\varphi}w) &= \mathfrak{Re}\left(e^{i\varphi}\sum_{j=1}^{k} y_j^* B y_j\right) \\
&= \sum_{j=1}^{k} y_j^* \mathfrak{Re}(e^{i\varphi}B) y_j \\
&= \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}B)) \\
&> \sum_{j=1}^{k} \lambda_j(\mathfrak{Re}(e^{i\varphi}A)).
\end{aligned}$$

Therefore, by Theorem 4.12, we have $W_k(B) \not\subseteq W_k(A)$. □

Chapter 5

Results on the Numerical Range of Cyclic Shift Matrices

We begin this chapter by considering the following problem:

Problem 1: Let $a_1, \ldots, a_n \in \mathbb{R}$ such that $0 \leq a_1 \leq \cdots \leq a_n$. Find $\sigma \in S_n$ such that $W\left(S\left(a_{\tau(1)}, \ldots, a_{\tau(n)}\right)\right) \subseteq W\left(S\left(a_{\sigma(1)}, \ldots, a_{\sigma(n)}\right)\right)$ for all $\tau \in S_n$.

In addition, we present the works of Chien et al. [4] and Gau [7] to highlight the methods and strategies used to solve Problem 1 and prove the following conjecture.

Conjecture 5.1. *The cyclic shift $S(a_1, a_3, a_5, \ldots, a_6, a_4, a_2)$ with $0 \leq a_1 \leq \cdots \leq a_n$ has the largest numerical range among all cyclic shift matrices with the same collection of weights.*

5.1 Results for $n \leq 8$

Let $A, B \in M_n(\mathbb{C})$. The authors first established a necessary and sufficient condition for the inclusion relation $W(B) \subseteq W(A)$ to hold. From Theorem 3.7, recall that

$$W(A) = \bigcap_{\theta \in [0, 2\pi)} \left\{ v \in \mathbb{C} : \mathfrak{Re}\left(e^{i\theta} v\right) \leq \lambda_1\left(\mathfrak{Re}\left(e^{i\theta} A\right)\right) \right\}.$$

By Corollary 4.13, $W(B) \subseteq W(A)$ if and only if $\lambda_1\left(\mathfrak{Re}\left(e^{i\theta} B\right)\right) \leq \lambda_1\left(\mathfrak{Re}\left(e^{i\theta} A\right)\right)$ for all $\theta \in [0, 2\pi)$. Since we are comparing the largest eigenvalues of two matrices, which can be viewed as roots of two characteristic polynomials, the following lemma is useful.

Lemma 5.2. *[4, Lemma 2.4] Let $f(t)$ and $g(t)$ be monic polynomials of degree n, each has n real zeros. If either*

(1) $g(t_1) \geq f(t_1)$, where t_1 is the largest zero of $g(t)$, or

(2) $g(t) > f(t)$ for $t \geq t_2$, where t_2 is the largest zero of $f(t)$,

then the largest zero of $f(t)$ is larger than or equal to that of $g(t)$.

5.1.1 Results for $n = 2, 3$

Theorem 5.3. *[4, Theorem 4.1] Let $A = S(a_1, a_2), B = S(b_1, b_2) \in M_2(\mathbb{C})$ be cyclic shift matrices with nonnegative real weights. Then $W(B) \subseteq W(A)$ if and only if $|b_1 - b_2| \leq |a_1 - a_2|$ and $b_1 + b_2 \leq a_1 + a_2$.*

Proof. By the elliptical range theorem, $W(A)$ is an elliptical disk with major axis $\left[-\frac{a_1+a_2}{2}, \frac{a_1+a_2}{2}\right]$ and minor axis $\left[-\frac{i|a_1-a_2|}{2}, \frac{i|a_1-a_2|}{2}\right]$. This implies that $W(B) \subseteq W(A)$ if and only if the major and minor axes of A contain the respective axes of B. The claim follows from this observation. □

Theorem 5.4. *[4, Theorem 4.2] Let $A = S(a_1, a_2, a_3), B = S(b_1, b_2, b_3) \in M_3(\mathbb{C})$ be cyclic shift matrices with nonnegative real weights. Then $W(B) \subseteq W(A)$ if and only if for all $\theta \in [0, 2\pi)$, the largest zero of $x^3 - (b_1^2 + b_2^2 + b_3^2)x - 2b_1b_2b_3\cos(3\theta)$ is at most the largest zero of $x^3 - (a_1^2 + a_2^2 + a_3^2)x - 2a_1a_2a_3\cos(3\theta)$. Consequently, we have the following:*

1. *Suppose $a_1a_2a_3 = b_1b_2b_3$. Then $W(B) \subseteq W(A)$ if and only if*

$$a_1^2 + a_2^2 + a_3^2 \geq b_1^2 + b_2^2 + b_3^2.$$

2. *Suppose $a_1^2 + a_2^2 + a_3^2 = b_1^2 + b_2^2 + b_3^2$. Then $W(B) \subseteq W(A)$ if and only if $a_1a_2a_3 \geq b_1b_2b_3$.*

3. *For any permutation $\tau \in S_3$, $W(A) = W(S(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}))$.*

Proof. Suppose $W(B) \subseteq W(A)$. Note that

$$p_\theta(x) = x^3 - (a_1^2 + a_2^2 + a_3^2)x - 2a_1a_2a_3\cos(3\theta)$$

and

$$q_\theta(x) = x^3 - (b_1^2 + b_2^2 + b_3^2)x - 2b_1b_2b_3\cos(3\theta)$$

are the characteristic polynomials of $2\mathfrak{Re}(e^{i\theta}A)$ and $2\mathfrak{Re}(e^{i\theta}B)$, respectively, for all $\theta \in [0, 2\pi)$. By Corollary 4.13, $W(B) \subseteq W(A)$ if and only if for all $\theta \in [0, 2\pi)$, the largest zero of $q_\theta(x)$ is at most the largest zero of $p_\theta(x)$. We then use Lemma 5.2 to establish the first part and the second part, while the third part is a consequence of either of the first two. □

5.1.2 Results for $n = 4, 5$

As emphasized at the beginning of Section 5.1, computing the characteristic polynomial of $2\mathfrak{Re}(e^{i\theta}A)$ for all $\theta \in [0, 2\pi)$ is essential to establish the inclusion relation $W(B) \subseteq W(A)$ for any cyclic shift matrices $A, B \in M_n(\mathbb{C})$ with the same collection of weights.

Lemma 5.5. *[4, Lemma 4.3] Let $A = S(a_1, a_2, a_3, a_4) \in M_4(\mathbb{C})$ be a cyclic shift with nonnegative real weights. Then for all $\theta \in [0, 2\pi)$,*

$$det(xI_4 - 2\mathfrak{Re}(e^{i\theta}A)) = x^4 - (a_1^2 + a_2^2 + a_3^2 + a_4^2)x^2 + a_1^2a_3^2 + a_2^2a_4^2 - 2a_1a_2a_3a_4\cos(4\theta).$$

Lemma 5.6. *[4, Lemma 5.1] Let $A = S(a_1, a_2, a_3, a_4, a_5) \in M_5(\mathbb{C})$ be a cyclic shift with nonnegative real weights. Then for all $\theta \in [0, 2\pi)$,*

$$det(xI_5 - 2\mathfrak{Re}(e^{i\theta}A)) = x^5 - \alpha_1 x^3 + \alpha_2 x - 2\alpha_3\cos(5\theta),$$

where

$$\begin{aligned}\alpha_1 &= a_1^2 + a_2^2 + a_3^2 + a_4^2 + a_5^2\\ \alpha_2 &= a_1^2a_3^2 + a_1^2a_4^2 + a_2^2a_4^2 + a_2^2a_5^2 + a_3^2a_5^2\\ \alpha_3 &= a_1a_2a_3a_4a_5.\end{aligned}$$

Theorem 5.7. *[4, Theorem 4.4] Let $A = S(a_1, a_2, a_3, a_4)$ and $B = S(b_1, b_2, b_3, b_4)$ be cyclic shift matrices with the same collection of nonnegative real weights. Then $W(B) \subseteq W(A)$ if and only if $a_1^2a_3^2 + a_2^2a_4^2 \leq b_1^2b_3^2 + b_2^2b_4^2$. In particular, if $a_1 \leq a_2 \leq a_3 \leq a_4$, then for any $\tau \in S_4$, we have*

$$W(S(a_1, a_3, a_2, a_4)) \subseteq W(S(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)})) \subseteq W(S(a_1, a_3, a_4, a_2)).$$

Theorem 5.8. *[4, Proposition 5.2, Theorem 5.3] Let $A = S(a_1, a_2, a_3, a_4, a_5), B = S(b_1, b_2, b_3, b_4, b_5) \in M_5(\mathbb{C})$ be cyclic shift matrices with the same collection of nonnegative real weights. Then $W(B) \subseteq W(A)$ if and only if*

$$a_1^2a_3^2 + a_1^2a_4^2 + a_2^2a_4^2 + a_2^2a_5^2 + a_3^2a_5^2 \leq b_1^2b_3^2 + b_1^2b_4^2 + b_2^2b_4^2 + b_2^2b_5^2 + b_3^2b_5^2.$$

In particular, if $a_1 \leq a_2 \leq a_3 \leq a_4 \leq a_5$, then for any $\tau \in S_5$, we have

$$\begin{aligned}W(S(a_1, a_4, a_3, a_2, a_5)) &\subseteq W(S(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)}, a_{\tau(5)}))\\ &\subseteq W(S(a_1, a_3, a_5, a_4, a_2)).\end{aligned}$$

5.1.3 Results for $n = 6$

Let $A = S(a_1, a_2, a_3, a_4, a_5, a_6), B = S(b_1, b_2, b_3, b_4, b_5, b_6) \in M_6(\mathbb{C})$ be cyclic shift matrices with the same collection of nonnegative real weights. Fix $\theta \in [0, 2\pi)$. By [4, Lemma 6.2], the characteristic polynomial of $2\mathfrak{Re}(e^{i\theta}A)$ is given by

$$p_\theta(x) = x^6 - \alpha_1 x^4 + \alpha_2 x^2 - \alpha_3 - 2\alpha_4 \cos(6\theta),$$

where

$$\begin{aligned}
\alpha_1 &= a_1^2 + a_2^2 + a_3^2 + a_4^2 + a_5^2 + a_6^2 \\
\alpha_2 &= a_1^2a_3^2 + a_1^2a_4^2 + a_1^2a_5^2 + a_2^2a_4^2 + a_2^2a_5^2 + a_2^2a_6^2 + a_3^2a_5^2 + a_3^2a_6^2 + a_4^2a_6^2 \\
\alpha_3 &= a_1^2a_3^2a_5^2 + a_2^2a_4^2a_6^2 \\
\alpha_4 &= a_1a_2a_3a_4a_5a_6.
\end{aligned}$$

Let $q_\theta(x) = x^6 - \beta_1 x^4 + \beta_2 x^2 - \beta_3 - 2\beta_4 \cos(6\theta)$ be the characteristic polynomial of $2\mathfrak{Re}(e^{i\theta}B)$. The comparison of the numerical ranges of A and B falls into two types: either $\alpha_3 = \beta_3$ or $\alpha_3 \neq \beta_3$. Since permutations are bijections, we have $\{\sigma(1), \sigma(3), \sigma(5)\} = \{\tau(1), \tau(3), \tau(5)\}$ if and only if $\{\sigma(2), \sigma(4), \sigma(6)\} = \{\tau(2), \tau(4), \tau(6)\}$ for any $\sigma, \tau \in S_6$. Hence, we divide the $\dfrac{6!}{2 \cdot 6} = 60$ representatives into 10 families, each of which has 6 members sharing the same value of α_3. From each family, we select one representative with the largest numerical range. Finally, we choose one from these 10 representatives with the largest numerical range. This leads us to the following theorem.

Theorem 5.9. *[4, Theorem 6.1] Let $a_j \in \mathbb{R}$ be nonnegative for all $j \in \{1, \ldots, 6\}$ such that $a_1 \leq a_2 \leq a_3 \leq a_4 \leq a_5 \leq a_6$. Then*

$$W(S(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)}, a_{\tau(5)})) \subseteq W(S(a_1, a_3, a_5, a_6, a_4, a_2))$$

for all $\tau \in S_6$.

For $n = 7$ or 8, the authors used a symbolic extension of the proof for $n = 6$ with the aid of Mathematica.

5.2 A Proof of the Conjecture

The following are some important properties of cyclic shift matrices.

Lemma 5.10. *[7, Proposition 1.1] Let $a_1, \ldots, a_n \in \mathbb{C}$. The following hold:*

1. *$S(a_1, \ldots, a_{n-r}, a_{n-r+1}, \ldots, a_n)$ is unitarily similar to $S(a_{1+r}, \ldots, a_n, a_1, \ldots, a_r)$ for $r \in \{1, \ldots, n-1\}$.*

2. *$S(a_1 e^{i\theta_1}, \ldots, a_n e^{i\theta_n})$ is unitarily similar to $S(a_1, \ldots, a_{n-1}, a_n e^{i\sum_{j=1}^{n}\theta_j})$ for all $\theta \in [0, 2\pi)$, where $a_j = |a_j| e^{i\theta_j}$ for $j \in \{1, \ldots, n\}$.*

3. *$W(S(a_1, a_2, \ldots, a_n)) = W(S(a_n, a_{n-1}, \ldots, a_1))$.*

Proof. For the first part, let $U = S(1, \ldots, 1) \in M_n(\mathbb{C})$. Then U is unitary and

$$U^r S(a_1, \ldots, a_n)(U^r)^* = S(a_{1+r}, \ldots, a_n, a_1, \ldots, a_r)$$

for $r \in \{1, \ldots, n-1\}$.

For the second part, let $U = \operatorname{diag}(1, e^{i\theta_1}, e^{i(\theta_1+\theta_2)}, \ldots, e^{i\sum_{j=1}^{n-1}\theta_j})$. Then U is unitary and

$$US(a_1 e^{i\theta_1}, \ldots, a_n e^{i\theta_n})U^* = S(a_1, \ldots, a_{n-1}, a_n e^{i\sum_{j=1}^{n}\theta_j}).$$

For the third part, note that $W(S(a_1, \ldots, a_n)) = W(S(a_1, \ldots, a_n)^\top)$ by definition, and

$$W(S(a_1, \ldots, a_n)^\top) = W(S(a_{n-1}, a_{n-2}, \ldots, a_1, a_n))$$

since $S(a_1, \ldots, a_n)^\top$ is unitarily similar to $S(a_{n-1}, a_{n-2}, \ldots, a_1, a_n)$ by the unitary matrix $U = \begin{bmatrix} & & 1 \\ & \iddots & \\ 1 & & \end{bmatrix}$. Moreover, the first part implies that

$$W(S(a_{n-1}, a_{n-2}, \ldots, a_1, a_n)) = W(S(a_n, a_{n-1}, \ldots, a_2, a_1)).$$

Combining the equalities gives us $W(S(a_1, \ldots, a_n)) = W(S(a_n, a_{n-1}, \ldots, a_1))$. □

In particular, by unitary similarity invariance, the k-numerical range of a cyclic shift is invariant under a *cyclic* or *reversal* permutation of its weights. Thus, we have the following theorem.

Theorem 5.11. *[7, Theorem 2.1] Let $a_1, \ldots, a_n \in \mathbb{R}$ such that $0 \leq a_n \leq \cdots \leq a_1$. Then*

$$W\left(S\left(a_{\tau(1)}, \ldots, a_{\tau(n)}\right)\right) \subseteq W\left(S\left(a_{\sigma(1)}, \ldots, a_{\sigma(n)}\right)\right)$$

for all $\tau \in S_n$, where σ is defined by

$$(\sigma(1), \ldots, \sigma(n)) = \begin{cases} (n-1, \ldots, 4, 2, 1, 3, 5, \ldots, n-2, n), & \text{if } n \text{ is odd} \\ (n-2, \ldots, 4, 2, 1, 3, 5, \ldots, n-1, n), & \text{if } n \text{ is even.} \end{cases}$$

We now provide a shortened proof of Theorem 5.11.

Proof. Let $A = S\left(a_{\sigma(1)}, \ldots, a_{\sigma(n)}\right)$ and $B = S(b_1, \ldots, b_n)$ be cyclic shift matrices with the same collection of nonnegative real weights such that $a_n \leq \cdots \leq a_1$. Fix $\theta \in [0, 2\pi)$. Without loss of generality, suppose $e^{i\theta}B = S(b_1, \ldots, b_{n-1}, b_n e^{in\theta})$. Fix a unit eigenvector $x = [x_1 \ \cdots \ x_n]^\top$ of $\mathfrak{Re}(e^{i\theta}B)$ corresponding to $\lambda_1(\mathfrak{Re}(e^{i\theta}B))$. For each $j \in \{1, \ldots, n\}$, write $x_j = |x_j|e^{i\theta_j}$ for some $\theta_j \in [0, 2\pi)$. If $x_j = 0$ for some j, let $\theta_j = 0$. Moreover, let $x_{n+1} = x_1$ and $\theta_{n+1} = \theta_1$. Define

$$r_j = \begin{cases} \cos(\theta_{j+1} - \theta_j), & \text{if } j \in \{1, \ldots, n-1\} \\ \cos(n\theta + \theta_1 - \theta_n), & \text{if } j = n \end{cases}$$

and let $\eta, \mu \in S_n$ such that $r_{\eta(1)} \geq \cdots \geq r_{\eta(n)}$ and $|x_{\mu(1)}| \geq \cdots \geq |x_{\mu(n)}|$. Let $\ell \in \{1, \ldots, n\}$ such that $\eta(\sigma(\ell)) = n$. Similar to $e^{i\theta}B$, we can assume that

$$e^{i\theta}A = S\left(a_{\sigma(1)}, \ldots, a_{\sigma(\ell-1)}, e^{in\theta}a_{\sigma(\ell)}, a_{\sigma(\ell+1)}, \ldots, a_{\sigma(n)}\right).$$

By taking

$$y = \begin{bmatrix} \vdots \\ \left|x_{\mu(2j+1)}\right| e^{it_{2j+1}} \\ \vdots \\ \left|x_{\mu(3)}\right| e^{it_3} \\ \left|x_{\mu(1)}\right| e^{it_1} \\ \left|x_{\mu(2)}\right| e^{it_2} \\ \vdots \\ \left|x_{\mu(2j)}\right| e^{it_{2j}} \\ \vdots \end{bmatrix} \in \mathbb{C}^n,$$

where

$$t_1 = 0,$$
$$t_{2j} = \sum_{i=1}^{j} (\theta_{\eta(2i-1)+1} - \theta_{\eta(2i-1)}) \text{ for } j \in \left\{1, \ldots, \left\lfloor \frac{n}{2} \right\rfloor\right\},$$

and

$$t_{2j+1} = -\sum_{i=1}^{j} (\theta_{\eta(2i)+1} - \theta_{\eta(2i)}) \text{ for } j \in \left\{1, \ldots, \left\lfloor \frac{n-1}{2} \right\rfloor\right\},$$

we have

$$\begin{aligned}
\lambda_1(\mathfrak{Re}(e^{i\theta}B)) &= x^*\mathfrak{Re}(e^{i\theta}B)x \\
&= \mathfrak{Re}(x^*(e^{i\theta}B)x) \\
&= \mathfrak{Re}\left(e^{in\theta}b_n x_1\overline{x_n} + \sum_{j=1}^{n-1} b_j\overline{x_j}x_{j+1}\right) \\
&= \sum_{j=1}^{n} b_j|x_j||x_{j+1}|r_j \\
&\leq y^*\mathfrak{Re}(e^{i\theta}A)y \\
&\leq \lambda_1(\mathfrak{Re}(e^{i\theta}A)),
\end{aligned}$$

where the first inequality holds due to [7, Lemma 2.5] while the last inequality holds due to Theorem 2.26. By Theorem 3.7, it follows that $W(B) \subseteq W(A)$. □

Gau (2024) extended this result to cyclic shift matrices whose weights are complex.

Theorem 5.12. *[7, Corollary 2.6] Let $a_1, \ldots, a_n \in \mathbb{C}$ such that $|a_n| \leq \cdots \leq |a_1|$. The cyclic shift $S(a_{\sigma(1)}, \ldots, a_{\sigma(n)})$, where $\sigma \in S_n$, as defined in Theorem 5.11, has the largest numerical range among all cyclic shift matrices with the same collection of weights.*

Proof. Let $A = S(a_1, a_3, a_5, \ldots, a_6, a_4, a_2)$ and $B = S(b_1, \ldots, b_n)$ be cyclic shift matrices with the same collection of weights. For $j \in \{1, \ldots, n\}$, write $a_j = |a_j|e^{i\theta_j}$ for some $\theta_j \in [0, 2\pi)$ and let $\theta = \dfrac{1}{n}\displaystyle\sum_{j=1}^{n}\theta_j$. Then

$$A = S(|a_1|e^{i\theta_1}, |a_3|e^{i\theta_3}, |a_5|e^{i\theta_5}, \ldots, |a_6|e^{i\theta_6}, |a_4|e^{i\theta_4}, |a_2|e^{i\theta_2}).$$

By Proposition 5.10, A is unitarily similar to

$$S(|a_1|, |a_3|, |a_5|, \ldots, |a_6|, |a_4|, |a_2|e^{i\sum_{j=1}^{n}\theta_j}) = S(|a_1|, |a_3|, |a_5|, \ldots, |a_6|, |a_4|, |a_2|e^{in\theta}),$$

which is unitarily similar to

$$S(|a_1|e^{i\theta}, |a_3|e^{i\theta}, \ldots, |a_4|e^{i\theta}, |a_2|e^{i\theta}) = e^{i\theta}S(|a_1|, |a_3|, |a_5|, \ldots, |a_6|, |a_4|, |a_2|).$$

Using the same reasoning, B is unitarily similar to $e^{i\theta}S(|b_1|, \ldots, |b_n|)$. By Theorem 5.11,

$$W(S(|b_1|, \ldots, |b_n|)) \subseteq W(S(|a_1|, |a_3|, |a_5|, \ldots, |a_6|, |a_4|, |a_2|)).$$

In particular, by Proposition 3.1,

$$\begin{aligned} W(B) &= W(e^{i\theta}S(|b_1|,\ldots,|b_n|)) \\ &= e^{i\theta}W(S(|b_1|,\ldots,|b_n|)) \\ &\subseteq e^{i\theta}W(S(|a_1|,|a_3|,|a_5|,\ldots,|a_6|,|a_4|,|a_2|)) \\ &= W(e^{i\theta}S(|a_1|,|a_3|,|a_5|,\ldots,|a_6|,|a_4|,|a_2|)) \\ &= W(A). \end{aligned}$$

□

Chapter 6

Results on the k-Numerical Range of Cyclic Shift Matrices

We formulate an analogue of Conjecture 5.1 for the k-numerical range.

Problem 2: Let $k \in \{1, \ldots, n\}$ and $a_1, \ldots, a_n \in \mathbb{R}$ such that $0 \leq a_1 \leq \cdots \leq a_n$. Find $\sigma \in S_n$ such that $W_k\left(S\left(a_{\tau(1)}, \ldots, a_{\tau(n)}\right)\right) \subseteq W_k\left(S\left(a_{\sigma(1)}, \ldots, a_{\sigma(n)}\right)\right)$ for all $\tau \in S_n$.

In this chapter, we solve Problem 2 for $n \in \{2, 3, 4\}$. We begin with the following two propositions.

Proposition 6.1. *Let $A \in M_n(\mathbb{C})$ be a cyclic shift. Then*

$$W_n(A) = \{0\}.$$

Proof. Since A is a cyclic shift, we have $\operatorname{tr}(A) = 0$. Thus, by the fifth part of Proposition 4.2, we get $W_n(A) = \{0\}$. □

Proposition 6.2. *Let $A, B \in M_n(\mathbb{C})$ be cyclic shift matrices. Then for any $k \in \{1, \ldots, n-1\}$, we have $W_k(B) \subseteq W_k(A)$ if and only if $W_{n-k}(B) \subseteq W_{n-k}(A)$.*

Proof. Let $k \in \{1, \ldots, n-1\}$.

Suppose $W_k(B) \subseteq W_k(A)$. By Proposition 4.3,

$$-W_{n-k}(B) = \operatorname{tr}(B) - W_{n-k}(B) = W_k(B) \subseteq W_k(A) = -W_{n-k}(A),$$

that is, $W_{n-k}(B) \subseteq W_{n-k}(A)$. The converse can be proven similarly. □

In the following sections, we omit the cases when $k = 1$ since these were already established by Chien et al. in [4] and by the fact that $W_1(A) = W(A)$.

6.1 Results for $n = 2, 3$

The following two theorems are immediate consequences of Proposition 6.1.

Theorem 6.3. *Let $A = S(a_1, a_2), B = S(b_1, b_2) \in M_2(\mathbb{C})$. Then $W_2(B) = W_2(A)$.*

Theorem 6.4. *Let $A = S(a_1, a_2, a_3), B = S(b_1, b_2, b_3) \in M_3(\mathbb{C})$. Then*

$$W_3(B) = W_3(A).$$

Proposition 6.2 tells us that for all cyclic shift matrices $A, B \in M_3(\mathbb{C})$, we have $W(B) \subseteq W(A)$ if and only if $W_2(B) \subseteq W_2(A)$. Thus, Theorem 5.4 gives us the following result.

Theorem 6.5. *Suppose $A = S(a_1, a_2, a_3)$ and $B = S(b_1, b_2, b_3)$ are cyclic shift matrices with nonnegative real weights. Then $W_2(B) \subseteq W_2(A)$ if and only if for $\theta \in [0, 2\pi)$, the largest zero of $x^3 - (b_1^2 + b_2^2 + b_3^2)x - 2b_1b_2b_3\cos(3\theta)$ is at most the largest zero of $x^3 - (a_1^2 + a_2^2 + a_3^2)x - 2a_1a_2a_3\cos(3\theta)$. Consequently, we have the following:*

1. *Suppose $a_1a_2a_3 = b_1b_2b_3$. Then $W_2(B) \subseteq W_2(A)$ if and only if*

$$a_1^2 + a_2^2 + a_3^2 \geq b_1^2 + b_2^2 + b_3^2.$$

2. *Suppose $a_1^2 + a_2^2 + a_3^2 = b_1^2 + b_2^2 + b_3^2$. Then $W_2(B) \subseteq W_2(A)$ if and only if $a_1a_2a_3 \geq b_1b_2b_3$.*

3. *For any permutation $\tau \in S_3$, $W_2(A) = W_2(S(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}))$.*

Theorems 5.4 and 6.5, together with Proposition 6.4, complete the case where $A \in M_3(\mathbb{C})$ is a cyclic shift with nonnegative real weights. By using a similar approach in Theorem 5.12, the following holds for cyclic shift matrices in $M_3(\mathbb{C})$ with complex weights.

Theorem 6.6. *Let $a_1, a_2, a_3 \in \mathbb{C}$. Then*

$$W_2(S(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)})) = W_2(S(a_1, a_3, a_2))$$

for any $\tau \in S_3$.

6.2 Results for $n = 4$

Let $A, B \in M_4(\mathbb{C})$ be cyclic shift matrices. Similar to the case where $n = 3$, we have $W_4(A) = W_4(B)$ and $W(B) \subseteq W(A)$ if and only if $W_3(B) \subseteq W_3(A)$. Thus, we are left with the case where $n = 4$ and $k = 2$. We begin with the following lemma.

Lemma 6.7. *Let $A = S(a_1, a_2, a_3, a_4) \in M_4(\mathbb{C})$ be a cyclic shift with nonnegative real weights. Then for all $\theta \in [0, 2\pi)$,*

$$\lambda_1\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) = \frac{1}{2}\left(\frac{1}{2}\alpha_1 + \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta))}\right)^{\frac{1}{2}},$$
$$\lambda_2\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) = \frac{1}{2}\left(\frac{1}{2}\alpha_1 - \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta))}\right)^{\frac{1}{2}},$$
$$\lambda_3\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) = -\frac{1}{2}\left(\frac{1}{2}\alpha_1 - \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta))}\right)^{\frac{1}{2}},$$

and

$$\lambda_4\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) = -\frac{1}{2}\left(\frac{1}{2}\alpha_1 + \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\cos(4\theta))}\right)^{\frac{1}{2}}$$

where

$$\alpha_1 = a_1^2 + a_2^2 + a_3^2 + a_4^2,$$
$$\alpha_2 = a_1^2a_3^2 + a_2^2a_4^2,$$
$$\alpha_3 = a_1a_2a_3a_4.$$

In addition,

$$\left(\lambda_1\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) + \lambda_2\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right)^2 = \frac{1}{4}\left(\alpha_1 + 2\sqrt{(\alpha_2 - 2\alpha_3\cos(4\theta))}\right).$$

Proof. Let $A = S(a_1, a_2, a_3, a_4) \in M_4(\mathbb{C})$ be a cyclic shift with nonnegative real weights. Fix $\theta \in [0, 2\pi)$. From Lemma 5.5, recall that

$$\det(xI_4 - 2\mathfrak{Re}(e^{i\theta}A)) = x^4 - \alpha_1x^2 + \alpha_2 - 2\alpha_3\cos(4\theta)$$

for all $\theta \in [0, 2\pi)$. If $z = x^2$, then $\det(xI_4 - 2\mathfrak{Re}(e^{i\theta}A)) = z^2 - \alpha_1z + \alpha_2 - 2\alpha_3\cos(4\theta)$. By the quadratic formula, the roots of $\det(xI_4 - 2\mathfrak{Re}(e^{i\theta}A))$ are

$$\lambda_1\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right) = \left(\frac{1}{2}\alpha_1 + \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta))}\right)^{\frac{1}{2}},$$
$$\lambda_2\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right) = \left(\frac{1}{2}\alpha_1 - \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta))}\right)^{\frac{1}{2}},$$
$$\lambda_3\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right) = -\left(\frac{1}{2}\alpha_1 - \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta))}\right)^{\frac{1}{2}},$$

and

$$\lambda_4\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right) = -\left(\frac{1}{2}\alpha_1 + \frac{1}{2}\sqrt{\alpha_1^2 - 4(\alpha_2 - 2\cos(4\theta))}\right)^{\frac{1}{2}}.$$

Notice that $\alpha_1 = \lambda_1\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right)^2 + \lambda_2\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right)^2$. Thus,

$$\begin{aligned}\left(\lambda_1\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) + \lambda_2\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right)^2 &= \left(\frac{1}{2}\lambda_1\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right) + \frac{1}{2}\lambda_2\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right)^2 \\ &= \frac{1}{4}\left(\alpha_1 + 2\sqrt{\frac{1}{4}\alpha_1^2 - \frac{1}{4}(\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta)))}\right) \\ &= \frac{1}{4}\left(\alpha_1 + \sqrt{\alpha_1^2 - (\alpha_1^2 - 4(\alpha_2 - 2\alpha_3\cos(4\theta)))}\right) \\ &= \frac{1}{4}\left(\alpha_1 + 2\sqrt{(\alpha_2 - 2\alpha_3\cos(4\theta))}\right).\end{aligned}$$

□

We use the preceding lemma to establish the following result.

Theorem 6.8. *Let $A = S(a_1, a_2, a_3, a_4)$ and $B = S(b_1, b_2, b_3, b_4)$ be cyclic shift matrices with nonnegative real weights. Let $\alpha_1 = a_1^2 + a_2^2 + a_3^2 + a_4^2$, $\alpha_2 = a_1^2a_3^2 + a_2^2a_4^2$, $\alpha_3 = a_1a_2a_3a_4$, $\beta_1 = b_1^2 + b_2^2 + b_3^2 + b_4^2$, $\beta_2 = b_1^2b_3^2 + b_2^2b_4^2$, and $\beta_3 = b_1b_2b_3b_4$. Then $W_2(A) \subseteq W_2(B)$ if and only if*

$$2\sqrt{\alpha_2 - 2\alpha_3\cos(4\theta)} + \alpha_1 \leq 2\sqrt{\beta_2 - 2\beta_3\cos(4\theta)} + \beta_1$$

for all $\theta \in [0, 2\pi)$.

Proof. By Corollary 4.13, it follows that $W_2(A) \subseteq W_2(B)$ if and only if

$$\sum_{j=1}^{2}\lambda_j\left(\mathfrak{Re}\left(e^{i\theta}A\right)\right) \leq \sum_{j=1}^{2}\lambda_j\left(\mathfrak{Re}\left(e^{i\theta}B\right)\right).$$

Recall that $\lambda_j(\mathfrak{Re}\left(e^{i\theta}A\right))$ and $\lambda_j(\mathfrak{Re}\left(e^{i\theta}B\right))$ are nonnegative for all $j \in \{1, 2\}$. Thus, by Lemma 6.7, the above inequality is equivalent to

$$\begin{aligned}2\sqrt{\alpha_2 - 2\alpha_3\cos(4\theta)} + \alpha_1 &= \left(\sum_{j=1}^{2}\lambda_j\left(2\mathfrak{Re}\left(e^{i\theta}A\right)\right)\right)^2 \\ &\leq \left(\sum_{j=1}^{2}\lambda_j\left(2\mathfrak{Re}\left(e^{i\theta}B\right)\right)\right)^2 \\ &= 2\sqrt{\beta_2 - 2\beta_3\cos(4\theta)} + \beta_1\end{aligned}$$

for all $\theta \in [0, 2\pi)$.

□

As a consequence, we have the following corollaries.

Corollary 6.9. *Let $A = S(a_1, a_2, a_3, a_4)$ and $B = S(b_1, b_2, b_3, b_4)$ be cyclic shift matrices with nonnegative real weights. Suppose $a_1^2 + a_2^2 + a_3^2 + a_4^2 = b_1^2 + b_2^2 + b_3^2 + b_4^2$ and $a_1a_2a_3a_4 = b_1b_2b_3b_4$. Then $W(B) \subseteq W(A)$ if and only if $W_2(A) \subseteq W_2(B)$.*

Proof. Let $\alpha_1 = a_1^2 + a_2^2 + a_3^2 + a_4^2$, $\alpha_2 = a_1^2a_3^2 + a_2^2a_4^2$, $\alpha_3 = a_1a_2a_3a_4$, $\beta_1 = b_1^2 + b_2^2 + b_3^2 + b_4^2$, $\beta_2 = b_1^2b_3^2 + b_2^2b_4^2$, and $\beta_3 = b_1b_2b_3b_4$. By assumption $\alpha_1 = \beta_1$ and so Theorem 6.8 gives us $W_2(A) \subseteq W_2(B)$ if and only if

$$2\sqrt{\alpha_2 - 2\alpha_3\cos(4\theta)} \leq 2\sqrt{\beta_2 - 2\beta_3\cos(4\theta)}$$

for all $\theta \in [0, 2\pi)$. Equivalently,

$$\alpha_2 - 2\alpha_3\cos(4\theta) \leq \beta_2 - 2\beta_3\cos(4\theta)$$

for all $\theta \in [0, 2\pi)$, that is, $\alpha_2 \leq \beta_2$, since $\alpha_3 = \beta_3$. By Theorem 5.7, this is equivalent to $W(B) \subseteq W(A)$. □

Corollary 6.10. *Let $A = S(a_1, a_2, a_3, a_4) \in M_4(\mathbb{C})$ be a cyclic shift with nonnegative real weights. Suppose $a_1 \leq a_2 \leq a_3 \leq a_4$. Then*

$$\begin{aligned} W_2(S(a_1, a_3, a_4, a_2)) &\subseteq W_2\left(S\left(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)}\right)\right) \\ &\subseteq W_2(S(a_1, a_3, a_2, a_4)) \end{aligned}$$

for any $\tau \in S_4$.

Proof. By Theorem 5.7,

$$\begin{aligned} W(S(a_1, a_3, a_2, a_4)) &\subseteq W\left(S\left(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)}\right)\right) \\ &\subseteq W(S(a_1, a_3, a_4, a_2)) \end{aligned}$$

Thus, by Corollary 6.9,

$$\begin{aligned} W_2(S(a_1, a_3, a_4, a_2)) &\subseteq W_2\left(S\left(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)}\right)\right) \\ &\subseteq W_2(S(a_1, a_3, a_2, a_4)). \end{aligned}$$

□

This completes the case where $n = 4$. Similar to the approach in Theorem 5.12, the following result holds for 4-by-4 cyclic shift matrices with complex weights.

Theorem 6.11. *Let $A = S(a_1, a_2, a_3, a_4) \in M_4(\mathbb{C})$, where $|a_1| \leq |a_2| \leq |a_3| \leq |a_4|$. Then*

$$\begin{aligned} W_2(S(a_1, a_3, a_4, a_2)) &\subseteq W_2\left(S\left(a_{\tau(1)}, a_{\tau(2)}, a_{\tau(3)}, a_{\tau(4)}\right)\right) \\ &\subseteq W_2(S(a_1, a_3, a_2, a_4)) \end{aligned}$$

for any $\tau \in S_4$.

Chapter 7

Summary and Recommendations

7.1 Summary

In [4], Chien et al. proposed a conjecture on the numerical range of cyclic shift matrices: the cyclic shift $S(a_1, a_3, a_5, \ldots, a_6, a_4, a_2)$ with $0 \leq a_1 \leq \cdots \leq a_n$ has the largest numerical range among all cyclic shift matrices with the same collection of weights. They proved the conjecture for $2 \leq n \leq 6$ and a symbolic extension of the proof of $n = 6$ was used for $n = 7, 8$. In [7], Gau proved the conjecture for all $n \in \mathbb{N}$.

In this study, we formulated a version of the problem for the k-numerical range as shown in Problem 2. For $n \in \{2, 3, 4\}$, the following were accomplished:

1. We established a necessary and sufficient condition for an n-by-n cyclic shift to have the largest 2-numerical range among all n-by-n cyclic shifts with the same collection of nonnegative real weights (Theorems 6.3, 6.5, and 6.8).

2. We proved that given weights $0 \leq a_1 \leq \cdots \leq a_n$, the cyclic shift $S(a_1, a_3, \ldots, a_2, a_4)$ has the largest 2-numerical range and that $S(a_1, a_3, \ldots, a_4, a_2)$ has the smallest 2-numerical range among all cyclic shift matrices with the same collection of weights.

3. We extended the above results to n-by-n cyclic shift matrices whose weights are complex as shown in Theorems 6.3, 6.6, and 6.11.

To establish the above results, we proved statements about the k-numerical range.

7.2 Recommendations

Let $n \in \mathbb{N}$ and $0 \leq a_1 \leq \cdots \leq a_n$ be given. Based on the summary, we recommend the following directions for future study.

1. Determine if $S(a_1, a_3, a_5, \ldots, a_2, a_4, a_6, \ldots)$ has the largest 2-numerical range among all cyclic shift matrices with the same collection of weights.

2. For $k \in \{2, \ldots, n\}$, find a permutation of weights that induces the largest k-numerical range among all cyclic shift matrices with the same collection of weights.

3. Formulate an analogous version of the conjecture for other generalizations of the numerical range. One might consider c-numerical ranges.

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